

COMPLEX ANALYSIS

Lecture Notes without Proofs

Notes from MATH 503, Fall 2025

Reading this edition

This edition follows the source order of the twenty-one lectures in *Complex Analysis Merged Notes(2).pdf*, MATH 503, Fall 2025. Source-slide numbers refer to pages of that merged PDF, not to the page numbers of this book. Definitions, statements, remarks, explanatory prose, and examples are kept; proofs are omitted. Explicit *Prove it*, *Why?*, and similar reader prompts are recorded as exercises.

The typesetting and numbering belong to this edition. Original illustrations are reproduced as vector figures. Recalled results are retained where they occur again. Editorial clarifications are identified separately. The accompanying coverage checklist records how every source page was handled.

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CHAPTER 1

Complex Numbers

MATH 503, Fall 2025. September 4, 2025. Merged PDF pages 1–27.

1.1 The complex field

Source slide 2.

1.1.1. Definition: *Complex numbers.* A complex number is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$.

1.1.2. Definition: *Addition and multiplication of complex numbers.* For two complex numbers $x = (a, b)$ and $y = (c, d)$, addition and multiplication are defined by

$$x + y = (a + c, b + d), \quad x \cdot y = (ac - bd, ad + bc).$$

Source slide 3.

1.1.3. Theorem (Complex field). — *These operations turn the set of all complex numbers into a field, with $(0, 0)$ and $(1, 0)$ playing respectively the roles of 0 and 1. This field is denoted by $\mathbb{C}$.*

Source slide 8.

1.1.4. Remark. For any $a, b \in \mathbb{R}$,

$$(a, 0) + (b, 0) = (a + b, 0), \quad (a, 0)(b, 0) = (ab, 0).$$

The complex numbers in $\{(a, 0) : a \in \mathbb{R}\}$ have the same arithmetic properties as the corresponding real numbers. We can therefore identify $(a, 0)$ with a . This identification gives the real field $\mathbb{R}$ as a subfield of $\mathbb{C}$.

We have defined $\mathbb{C}$ without reference to the mysterious square root of -1 . We now show that (a, b) is equivalent to the customary $a + bi$.

1.1.5. Definition. The imaginary number is $i = (0, 1)$.

Source slide 9.

1.1.6. Theorem. — *One has $i^2 = -1$.*

1.1.7. Theorem. — *One has $\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$.*

Source slide 10.

1.1.8. Definition: Conjugate, real and imaginary parts. For $z = a + ib$ with $a, b \in \mathbb{R}$, the number $\bar{z} = a - ib$ is its conjugate. The numbers a, b are respectively its real and imaginary parts:

$$a = \Re(z) = \operatorname{Re}(z), \quad b = \Im(z) = \operatorname{Im}(z).$$

1.1.9. Theorem. — For $z, w \in \mathbb{C}$:

- (i) $\overline{z + w} = \bar{z} + \bar{w}$.
- (ii) $\overline{zw} = \bar{z}\bar{w}$.
- (iii) $z + \bar{z} = 2\operatorname{Re} z$ and $z - \bar{z} = 2i\operatorname{Im} z$.
- (iv) $z\bar{z}$ is a positive real number except when $z = 0$.

Source slide 12.

1.1.10. Definition: Absolute value. For $z \in \mathbb{C}$, its absolute value is $|z| = \sqrt{z\bar{z}}$.

1.1.11. Remark. This absolute value exists and is unique. It coincides with the real absolute value: if $x \in \mathbb{R}$, then $\bar{x} = x$, and

$$|x| = \sqrt{x\bar{x}} = \sqrt{x^2} = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Source slide 13.

1.1.12. Theorem (Properties of absolute value). — For $z, w \in \mathbb{C}$:

- (i) $|z| > 0$ if and only if $z \neq 0$, and $|0| = 0$.
- (ii) $|\bar{z}| = |z|$.
- (iii) $|zw| = |z||w|$.
- (iv) $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$.
- (v) $|z + w| \leq |z| + |w|$.

Source slide 15.

1.1.13. Definition: Argument. An argument $\arg z$ of $z = a + ib$ is the angle which the segment from $(0, 0)$ to (a, b) makes with the positive real axis. The argument is not unique but is determined up to a multiple of 2π .

If $r = |z|$ and $\theta = \arg z$, then

$$z = r(\cos \theta + i \sin \theta).$$

Trigonometric identities give

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2, \quad \arg(z_1/z_2) = \arg z_1 - \arg z_2, \quad \arg \bar{z} = -\arg z.$$

An argument is *principal* if it lies in $(-\pi, \pi]$.

Editorial clarification. Arguments are defined only for nonzero numbers. The displayed argument equalities are understood modulo 2π .

1.2 Topology of the complex plane

Source slide 16.

1.2.1. Definition: Convergence. A sequence $(z_n)_{n \in \mathbb{N}} \subseteq \mathbb{C}$ converges to z , written $\lim_{n \rightarrow \infty} z_n = z$, if and only if $\lim_{n \rightarrow \infty} |z_n - z| = 0$. Equivalently, for every $\varepsilon > 0$ there is $N_\varepsilon \in \mathbb{N}$ such that $n \geq N_\varepsilon$ implies $|z_n - z| < \varepsilon$.

One has $z_n \rightarrow z$ if and only if $\operatorname{Re} z_n \rightarrow \operatorname{Re} z$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} z$. We write $z_n \rightarrow \infty$ when $|z_n| \rightarrow \infty$.

Source slide 17.

1.2.2. Definition: Cauchy sequence. A sequence (z_n) is Cauchy if $\lim_{m, n \rightarrow \infty} |z_n - z_m| = 0$. Equivalently, for every $\varepsilon > 0$ there is N_ε such that $m, n \geq N_\varepsilon$ implies $|z_n - z_m| < \varepsilon$.

1.2.3. Theorem. — The complex plane with metric $d(z_1, z_2) = |z_1 - z_2|$ is a complete metric space.

Source slide 18.

1.2.4. Definition: Discs, punctured discs, and circles. For $a \in \mathbb{C}$ and $r > 0$,

$$\begin{aligned} D(a, r) &= \{z : |z - a| < r\}, & \overline{D(a, r)} &= \{z : |z - a| \leq r\}, \\ D'(a, r) &= \{z : 0 < |z - a| < r\}, & C(a, r) &= \{z : |z - a| = r\}. \end{aligned}$$

The unit disc is $\mathbb{D} = D(0, 1)$. The punctured disc is open, and $C(a, r) = \overline{D(a, r)} \setminus D(a, r)$.

Source slide 19.

A set $\Omega \subseteq \mathbb{C}$ is *open* if each $a \in \Omega$ has a disc $D(a, r) \subseteq \Omega$. It is *closed* if its complement $\Omega^c = \mathbb{C} \setminus \Omega$ is open.

A point z is a *limit point* of Ω if there is a sequence $(z_n) \subseteq \Omega$ with $z_n \neq z$ and $z_n \rightarrow z$. A set is closed if and only if it contains all its limit points. Its closure $\overline{\Omega}$ is the union of the set and its limit points. The boundary $\partial\Omega$ is its closure minus its interior. For instance, $C(a, r) = \partial D(a, r)$.

Source slide 20.

For $a, b \in \mathbb{C}$, $[a, b]$ denotes the closed segment with endpoints a, b . For arbitrary $t_1 < t_2$,

$$[a, b] = \left\{ a + \frac{t - t_1}{t_2 - t_1} (b - a) : t_1 \leq t \leq t_2 \right\}.$$

For $a_1, \dots, a_{n+1} \in \mathbb{C}$, a *polygon* joining a_1 to a_{n+1} is $\bigcup_{j=1}^n [a_j, a_{j+1}]$, abbreviated $[a_1, \dots, a_{n+1}]$.

Source slide 21.

1.2.5. Definition: Connectedness. Let (X, d) be a metric space and $E \subseteq X$. The set E is connected if it cannot be written as a disjoint union of two nonempty relatively open subsets of E .

Thus $E = A \cup B$, $A \cap B = \emptyset$, with A, B open in E , implies $A = \emptyset$ or $B = \emptyset$. Otherwise this is called a *separation* of E into open sets. The union of two disjoint open discs A, B is not connected, since

$$E = A \cup B = (A \cap E) \cup (B \cap E),$$

where the two sets are nonempty, disjoint, and relatively open. An open connected set is called a *region*.

Source slide 22.

1.2.6. Definition: Component. A maximal connected subset of E is a component of E .

For $a \in E$, let $C(a)$ be the union of all connected subsets of E containing a . Since $\{a\}$ is connected, $a \in C(a)$, and $E = \bigcup_{a \in E} C(a)$.

- 1.2.7. Lemma.** — (i) $C(a)$ is connected.
(ii) Components of E are disjoint or identical.
(iii) Components of an open set are open.

1.2.8. Theorem. — An open set in the complex plane is a disjoint union of regions.

Editorial clarification. The slide states the last two openness assertions for arbitrary metric spaces. They require local connectedness; here they are applied to $\mathbb{C}$, where they hold.

Source slide 25.

1.2.9. Theorem (Polygonal connectedness). — A nonempty open subset E of $\mathbb{C}$ is connected if and only if any two points of E can be joined by a polygonal path lying in E .

CHAPTER 2

Compactification, Holomorphic Functions, and Cauchy–Riemann Equations

MATH 503, Fall 2025. September 8, 2025. Merged PDF pages 28–59.

2.1 The Riemann sphere

Source slide 29.

Let (X, d) be a metric space and $\Omega \subseteq X$. An open covering of Ω is a family $(U_\alpha)_{\alpha \in A}$ of open sets such that $\Omega \subseteq \bigcup_{\alpha \in A} U_\alpha$. The index set need not be countable.

2.1.1. Definition: Compactness. A set in a metric space is compact if and only if every open covering has a finite subcovering.

2.1.2. Theorem. — A set Ω in a metric space is compact if and only if every sequence in Ω has a subsequence converging to a point of Ω .

2.1.3. Theorem. — A subset of $\mathbb{C}$ is compact if and only if it is closed and bounded.

Source slide 30.

Adjoining a point ∞ , called the point at infinity, gives the *extended complex plane* $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$. The Riemann sphere is

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}.$$

The *stereographic projection* $f : S \rightarrow \mathbb{C}_\infty$ is

$$f(x_1, x_2, x_3) = \frac{x_1 + ix_2}{1 - x_3} \quad ((x_1, x_2, x_3) \neq (0, 0, 1)), \quad f(0, 0, 1) = \infty.$$

The point $N = (0, 0, 1)$ is the north pole. The map is one-to-one and onto.

Source slide 33–35.

Identify $z = x + iy$ with $(x, y, 0) \in \mathbb{R}^3$. The line joining z to N is

$$tN + (1 - t)z = ((1 - t)x, (1 - t)y, t), \quad -\infty < t < \infty.$$

Its intersections with S satisfy

$$(1 - t)^2 x^2 + (1 - t)^2 y^2 + t^2 = 1.$$

Away from the north pole, this becomes $(1 - t)|z|^2 = 1 + t$, hence $t = (|z|^2 - 1)/(|z|^2 + 1)$. The other intersection is

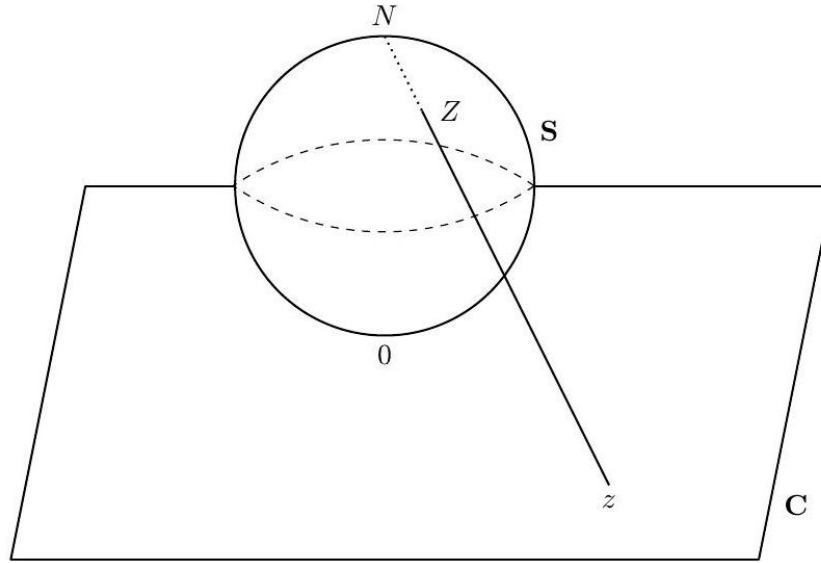


FIGURE 1. The stereographic projection diagram from source slide 34.

$$\begin{aligned}
 Z &= \left(\left[1 - \frac{|z|^2 - 1}{|z|^2 + 1} \right] x, \left[1 - \frac{|z|^2 - 1}{|z|^2 + 1} \right] y, \frac{|z|^2 - 1}{|z|^2 + 1} \right) \\
 &= \left(\frac{2x}{|z|^2 + 1}, \frac{2y}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1} \right) \\
 &= \left(\frac{z + \bar{z}}{|z|^2 + 1}, \frac{z - \bar{z}}{i(|z|^2 + 1)}, \frac{|z|^2 - 1}{|z|^2 + 1} \right).
 \end{aligned}$$

Thus the line through $(x, y, 0)$ and $(0, 0, 1)$ meets $S \setminus \{N\}$ in exactly one point $Z = (x_1, x_2, x_3)$, and $f(Z) = z$. These are the *spherical coordinates* of z ; those of ∞ are $(0, 0, 1)$.

Source slide 36–38.

For $z, z' \in \mathbb{C}_\infty$, let $x = (x_1, x_2, x_3)$ and $x' = (x'_1, x'_2, x'_3)$ be their spherical coordinates, as given above for each finite point. Define

$$d(z, z') = \sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + (x_3 - x'_3)^2}.$$

Consequently

$$d(z, z')^2 = 2 - 2(x_1 x'_1 + x_2 x'_2 + x_3 x'_3).$$

Writing $D = (|z|^2 + 1)(|z'|^2 + 1)$, the scalar product calculation is

$$\begin{aligned}
 x_1 x'_1 + x_2 x'_2 + x_3 x'_3 &= \frac{(z + \bar{z})(z' + \bar{z}') - (z - \bar{z})(z' - \bar{z}') + (|z|^2 - 1)(|z'|^2 - 1)}{D} \\
 &= \frac{(|z|^2 - 1)(|z'|^2 - 1) + 2z\bar{z}' + 2\bar{z}z'}{D} \\
 &= \frac{D - 2(|z|^2 + 1) - 2(|z'|^2 + 1) + 4 + 2z\bar{z}' + 2\bar{z}z'}{D}.
 \end{aligned}$$

The numerator subtracted from D is

$$\begin{aligned} & 2(|z|^2 + 1) + 2(|z'|^2 + 1) - 4 - 2z\bar{z}' - 2\bar{z}z' \\ &= 2(|z|^2 + |z'|^2 - z\bar{z}' - \bar{z}z') = 2|z - z'|^2. \end{aligned}$$

Therefore

$$2(x_1x'_1 + x_2x'_2 + x_3x'_3) = 2 - \frac{4|z - z'|^2}{D}, \quad d(z, z')^2 = \frac{4|z - z'|^2}{D},$$

$$d(z, z') = \frac{2|z - z'|}{\sqrt{(|z|^2 + 1)(|z'|^2 + 1)}}.$$

If $z' = \infty$, then $x'_1 = x'_2 = 0$, $x'_3 = 1$, and

$$d(z, \infty)^2 = 2 - 2x_3 = 2 - 2\frac{|z|^2 - 1}{|z|^2 + 1} = \frac{4}{|z|^2 + 1}.$$

Source slide 39.

Hence $d(z, \infty) = 2/\sqrt{|z|^2 + 1}$, and $\mathbb{C}_\infty$ is a metric space. For $E \subseteq \mathbb{C}_\infty$, the induced topology is characterized as follows:

- If $\infty \notin E$, then E is open in $\mathbb{C}_\infty$ if and only if it is open in $\mathbb{C}$.
- If $\infty \in E$, then E is open if and only if its complement in $\mathbb{C}_\infty$ is a closed bounded (equivalently compact) subset of $\mathbb{C}$.

Checking continuity of f and f^{-1} gives:

2.1.4. Theorem. — *The extended complex plane is homeomorphic to S . In particular, it is compact.*

2.2 Complex functions

Source slide 40.

A function f on $E \subseteq \mathbb{C}$ assigns to each $z \in E$ a unique complex number $w = f(z)$, its value at z . The set E is the domain of definition. If $z = x + iy$ and $w = u + iv = f(x + iy)$, then u, v depend on the real variables x, y , and

$$f(z) = u(x, y) + iv(x, y), \quad u = \operatorname{Re} f, \quad v = \operatorname{Im} f.$$

Source slide 41.

2.2.1. Example. For $f(z) = z^2$, $f(x + iy) = x^2 - y^2 + 2ixy$, so $\operatorname{Re} f = x^2 - y^2$ and $\operatorname{Im} f = 2xy$. Its domain is all of $\mathbb{C}$.

2.2.2. Example: Branches of roots. Assigning all n th roots to each complex number does not give a function in the preceding single-valued sense: each $z \neq 0$ has n roots. If $z = |z|(\cos \theta + i \sin \theta)$ with $\theta \in (-\pi, \pi]$, they are

$$z_k = \sqrt[n]{|z|} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right), \quad k = 0, \dots, n-1.$$

These determine n branches. Choosing a particular k makes a single-valued function. The $k = 0$ branch is the principal branch.

Source slide 42.

2.2.3. Definition: Continuity. For $f : E \rightarrow \mathbb{C}$ and $z_0 \in E$, continuity at z_0 means that, for every $\varepsilon > 0$, there is $\delta > 0$ such that $z \in E$ and $|z - z_0| < \delta$ imply $|f(z) - f(z_0)| < \varepsilon$. Equivalently, whenever $z_n \in E$ and $z_n \rightarrow z_0$, one has $f(z_n) \rightarrow f(z_0)$.

Continuity on E means continuity at every point. Sums and products of continuous functions are continuous. A complex function is continuous if and only if both its real and imaginary parts are continuous.

Source slide 43.

Convergence in $\mathbb{C}$ and in $\mathbb{R}^2$ is the same, so continuity as a complex function is equivalent to continuity as a function of x, y . The triangle inequality shows that continuity of f implies continuity of $|f|$. Here attaining a maximum at z_0 means $|f(z)| \leq |f(z_0)|$ for all $z \in E$, with the inequality reversed for a minimum.

2.2.4. Theorem. — *A continuous complex function on a compact set is bounded and attains a maximum and a minimum of its absolute value.*

Source slide 44.

2.2.5. Example. The function z^2 is continuous on $\mathbb{C}$. The principal square root is continuous on $\mathbb{C} \setminus (-\infty, 0]$. Recall its two values

$$z_k = \sqrt{|z|} \left(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2} \right), \quad k = 0, 1.$$

For $z = x + iy$ with fixed $x < 0$, as $y \rightarrow 0^+$ the argument tends to π and the principal root tends to $i\sqrt{|x|}$; as $y \rightarrow 0^-$, the argument tends to $-\pi$ and the root tends to $-i\sqrt{|x|}$. This exhibits the discontinuity across the negative real axis.

Editorial clarification. At $x = 0$ both limits are 0; the discontinuity concerns $x < 0$. This specifies the endpoint qualification in the source's $x \leq 0$ wording.

2.3 Complex differentiation

Source slide 45.

2.3.1. Definition. For $f : \Omega \rightarrow \mathbb{C}$ on an open set, the derivative at $z_0 \in \Omega$ is

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

if this limit exists. Explicitly, every $\varepsilon > 0$ must admit $\delta > 0$ such that

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon \quad (z \in D'(z_0, \delta)).$$

If this holds at every point, f is holomorphic or analytic on Ω . The class is denoted by $H(\Omega)$. A function holomorphic on all of $\mathbb{C}$ is entire.

Source slide 46.

If $f, g \in H(\Omega)$, then $f + g, fg \in H(\Omega)$, so $H(\Omega)$ is a ring. If g never vanishes, then f/g is holomorphic. For $\alpha, \beta \in \mathbb{C}$,

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \quad (fg)' = f'g + fg', \quad (f/g)' = \frac{f'g - fg'}{g^2}.$$

Compositions of holomorphic functions are also holomorphic.

Source slide 47.

2.3.2. Lemma (Chain rule). — If $f \in H(\Omega)$, $f(\Omega) \subseteq \Omega_1$, and $g \in H(\Omega_1)$, then $h = g \circ f \in H(\Omega)$ and $h'(z_0) = g'(f(z_0))f'(z_0)$.

Source slide 48.

2.3.3. Theorem. — Let g be analytic on an open set Ω_1 and f continuous on an open set Ω . Suppose $f(\Omega) \subseteq \Omega_1$, g' never vanishes, and $g(f(z)) = z$ for every $z \in \Omega$ (so f is one-to-one). Then f is analytic and $f' = 1/(g' \circ f)$.

Source slide 49.

2.3.4. Example. The function z is holomorphic, with derivative 1 . Every polynomial

$$p(z) = a_0 + a_1 z + \cdots + a_n z^n, \quad p'(z) = a_1 + 2a_2 z + \cdots + na_n z^{n-1},$$

is entire. The function $1/z$ is holomorphic on every open set not containing 0 , with derivative $-1/z^2$. By contrast, $f(z) = \bar{z}$ is not holomorphic:

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\bar{h}}{h},$$

which has different limits along real and purely imaginary $h \rightarrow 0$.

2.4 Real differentiability and the Cauchy–Riemann equations

Source slide 50.

2.4.1. Definition. For $E \subseteq \mathbb{R}^n$ open and $F : E \rightarrow \mathbb{R}^m$, differentiability at x means that there is a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{h \rightarrow 0} \frac{|F(x + h) - F(x) - Ah|}{|h|} = 0.$$

We write $A = F'(x)$. Differentiability in E means differentiability at every point. Equivalently,

$$F(x + h) - F(x) = F'(x)h + r(h), \quad \lim_{h \rightarrow 0} \frac{|r(h)|}{|h|} = 0.$$

Source slide 51.

For $F = (F_1, \dots, F_m)$, the derivative matrix, when it exists, is

$$F'(x_0) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1}(x_0) & \cdots & \frac{\partial F_1}{\partial x_n}(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1}(x_0) & \cdots & \frac{\partial F_m}{\partial x_n}(x_0) \end{bmatrix}.$$

If all partial derivatives exist near x_0 and are continuous at x_0 , then F is differentiable there. Existence of partial derivatives alone does not suffice: the function

$$F(x, y) = \begin{cases} y^3/(x^2 + y^2), & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

has both partial derivatives at the origin but is not differentiable there; its partial derivatives are not continuous there.

Editorial clarification. The slide introduces this example as a failed converse. The example specifically shows that existence of partial derivatives is insufficient; it is not an example of a differentiable function with discontinuous partial derivatives.

Source slide 52.

Recall that $\bar{z}$ is not complex differentiable because its difference quotient is $\bar{h}/h$. As a real-variable map it is $F(x, y) = (x, -y)$, which is indefinitely differentiable. Thus real differentiability does not guarantee holomorphicity. More generally, to $f = u + iv$ we associate $F(x, y) = (u(x, y), v(x, y))$.

Source slide 53.

For $h = (h_1, h_2) \rightarrow 0$, real differentiability gives

$$u(x_0 + h_1, y_0 + h_2) - u(x_0, y_0) = u_x(x_0, y_0)h_1 + u_y(x_0, y_0)h_2 + r_1(h),$$

$$v(x_0 + h_1, y_0 + h_2) - v(x_0, y_0) = v_x(x_0, y_0)h_1 + v_y(x_0, y_0)h_2 + r_2(h),$$

where $r(h) = (r_1(h), r_2(h))$ and $|r_j(h)|/|h| \rightarrow 0$ for $j = 1, 2$.

Source slide 54.

2.4.2. Theorem (Cauchy–Riemann equations). — If $f : \Omega \rightarrow \mathbb{C}$ is holomorphic, then

$$\frac{\partial f}{\partial x} = \frac{1}{i} \frac{\partial f}{\partial y}.$$

Writing $f = u + iv$ with u, v real valued gives $u_x = v_y$ and $u_y = -v_x$. These are the Cauchy–Riemann equations.

Source slide 57.

2.4.3. Theorem. — If $f = u + iv$ on an open Ω , and u, v are real differentiable and satisfy the Cauchy–Riemann equations throughout Ω , then f is holomorphic.

Source slide 59.

2.4.4. Example. The real differentiability hypothesis is essential. Consider $f(x, y) = \sqrt{|xy|}$ as a complex function with zero imaginary part. It has both partial derivatives at $(0, 0)$ and satisfies the Cauchy–Riemann equations there. It is not real differentiable at the origin, however, and is not complex differentiable there.

CHAPTER 3

Power Series and Integration over Curves

MATH 503, Fall 2025. September 11, 2025. Merged PDF pages 60–86.

3.1 Power series

Source slide 61.

Given $(w_n)_{n \geq 0} \subseteq \mathbb{C}$, the series $\sum_{n=0}^{\infty} w_n$ converges to w if $\lim_{n \rightarrow \infty} \sum_{k=0}^n w_k = w$; otherwise it diverges. It converges if and only if its partial sums are Cauchy, that is, $\lim_{m, n \rightarrow \infty} \sum_{k=m}^n w_k = 0$. It converges *absolutely* if $\sum |w_n|$ converges. As for real series, absolute convergence implies convergence. Absolute convergence is equivalent to boundedness of the partial sums $\sum_{k=0}^n |w_k|$.

Source slide 62.

3.1.1. Theorem (Ratio test). — For a series $\sum w_n$ of nonzero terms, if $\limsup_{n \rightarrow \infty} |w_{n+1}/w_n| < 1$, it converges absolutely. If $|w_{n+1}/w_n| \geq 1$ for all sufficiently large n , it diverges.

3.1.A. EXERCISE. Prove the ratio test.

3.1.2. Theorem (Root test). — For any complex series $\sum w_n$, if $\limsup_{n \rightarrow \infty} |w_n|^{1/n} < 1$, it converges absolutely. If this limsup is greater than 1, it diverges.

3.1.B. EXERCISE. Prove the root test.

Source slide 63.

For complex-valued functions (f_n) on a set S , pointwise convergence is equivalent to the pointwise Cauchy condition. Uniform convergence is equivalent to the uniform Cauchy condition: $|f_n(z) - f_m(z)| \rightarrow 0$ uniformly for $z \in S$ as $m, n \rightarrow \infty$.

3.1.3. Theorem (Weierstrass M-test). — If $|g_n(z)| \leq M_n$ on $S \subseteq \mathbb{C}$ and $\sum_{n=1}^{\infty} M_n < \infty$, then $\sum_{n=1}^{\infty} g_n(z)$ converges uniformly on S .

Source slide 64.

Series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$, with $a_n, z_0 \in \mathbb{C}$, are called *power series*. They are series of functions of the special form $f_n(z) = a_n(z - z_0)^n$.

3.1.4. Theorem. — If such a series converges at z with $|z - z_0| = r$, it converges absolutely in $D(z_0, r)$, uniformly on every closed subdisc and hence on every compact subset of $D(z_0, r)$.

Source slide 65.

3.1.5. Theorem (Radius of convergence). — The radius of convergence of $\sum a_n(z - z_0)^n$ is

$$r = \left(\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \right)^{-1},$$

with conventions $1/0 = \infty$ and $1/\infty = 0$. The series converges absolutely on $D(z_0, r)$, uniformly on compact subsets, and diverges when $|z - z_0| > r$.

Source slide 66.

3.1.6. Definition. A function $f : \Omega \rightarrow \mathbb{C}$ on an open set is representable by power series in Ω if, for every disc $D(a, r) \subseteq \Omega$, there are coefficients c_n with

$$f(z) = \sum_{n=0}^{\infty} c_n(z - a)^n \quad (z \in D(a, r)).$$

3.1.7. Theorem. — Such a function is holomorphic, and its derivative is also representable by power series. On the same disc,

$$f'(z) = \sum_{n=1}^{\infty} n c_n(z - a)^{n-1}.$$

Source slide 70.

3.1.8. Remark. The theorem applies again to f' , so derivatives of every order exist and are representable by power series:

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) c_n(z - a)^{n-k}.$$

Consequently $k!c_k = f^{(k)}(a)$ for $k = 0, 1, 2, \dots$, which ensures the uniqueness of the coefficient sequence at each center.

Source slide 71.

3.1.9. Example: Geometric series. The series $\sum_{n=0}^{\infty} z^n$ converges absolutely exactly for $|z| < 1$. Its sum there is $1/(1 - z)$, although that function is holomorphic on $\mathbb{C} \setminus \{1\}$. As in the real case,

$$\sum_{n=0}^N z^n = \frac{1 - z^{N+1}}{1 - z}, \quad z^{N+1} \rightarrow 0 \quad (|z| < 1).$$

Termwise differentiation gives, on $D(0, 1)$,

$$\frac{1}{(1 - z)^2} = \left(\frac{1}{1 - z} \right)' = \sum_{n=1}^{\infty} n z^{n-1}.$$

Source slide 72–75.

3.1.10. Example: Exponential and trigonometric functions. The complex exponential, cosine, and sine are defined by

$$e^z = \exp z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

They agree with the usual real functions on $\mathbb{R}$, and all three series converge absolutely on $\mathbb{C}$. Indeed $|z^n/n!| = |z|^n/n!$, so the exponential series is bounded in absolute value by $\sum |z|^n/n! = e^{|z|} < \infty$. This also proves uniform convergence on each bounded disc. The same argument applies to sine and cosine.

$$\exp'(z) = \sum_{n=1}^{\infty} n \frac{z^{n-1}}{n!} = \sum_{m=0}^{\infty} \frac{z^m}{m!} = \exp z, \quad \cos' z = -\sin z, \quad \sin' z = \cos z.$$

These functions are entire. Absolute convergence permits multiplication:

$$\begin{aligned} \left(\sum_{k=0}^{\infty} \frac{z^k}{k!} \right) \left(\sum_{m=0}^{\infty} \frac{w^m}{m!} \right) &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \frac{n!}{k!(n-k)!} z^k w^{n-k} \\ &= \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!}. \end{aligned}$$

Thus $\exp z \exp w = \exp(z+w)$, and a direct calculation gives $\exp(iy) = \cos y + i \sin y$ for real y . Moreover $\exp z = 1$ if and only if $z = 2k\pi i$, $k \in \mathbb{Z}$. For $z = x + iy$, the modulus condition and the trigonometric identity force $x = 0$; then $\cos y = 1$ and $\sin y = 0$ force $y = 2k\pi$. Consequently the complex exponential has period $2\pi i$: $\exp(z + 2k\pi i) = \exp z$ for $z \in \mathbb{C}$, $k \in \mathbb{Z}$.

3.2 Integration over paths

Source slide 76.

3.2.1. Definition: Curve. In a topological space X , a curve is a continuous map $\gamma : [\alpha, \beta] \rightarrow X$, where $\alpha < \beta$ and the interval is compact. Its parameter interval is $[\alpha, \beta]$ and its range is $\gamma^* = \gamma([\alpha, \beta]) = \{\gamma(t) : t \in [\alpha, \beta]\}$.

The range is compact and connected. The curve is closed if its initial point $\gamma(\alpha)$ equals its endpoint $\gamma(\beta)$. We distinguish the oriented geometric object γ^* from its parametrization γ ; the parametrization is not unique.

Source slide 77.

Two parametrizations $\gamma : [\alpha, \beta] \rightarrow \mathbb{C}$ and $\tilde{\gamma} : [\alpha_1, \beta_1] \rightarrow \mathbb{C}$ are equivalent if there is a continuously differentiable bijection $\phi : [\alpha_1, \beta_1] \rightarrow [\alpha, \beta]$ with $\phi' > 0$ and $\tilde{\gamma}(s) = \gamma(\phi(s))$. The positive derivative preserves orientation as s travels from α_1 to β_1 . Equivalent parametrizations give the same oriented curve, whose orientation is determined by increasing parameter.

Source slide 78.

3.2.2. Definition: Path. A path is a piecewise continuously differentiable plane curve. More precisely, it is a continuous complex function γ on $[\alpha, \beta]$ for which

there are finitely many points $\alpha = s_0 < s_1 < \cdots < s_n = \beta$, and γ has a continuous derivative on each $[s_{j-1}, s_j]$. Left and right derivatives at the intervening endpoints may differ. A closed path is a closed curve which is a path.

Source slide 79.

3.2.3. Definition: Path integral. If f is continuous on the trace of a path γ , define

$$\int_{\gamma} f(z) dz = \int_{\alpha}^{\beta} f(\gamma(t)) \gamma'(t) dt.$$

The integral on the right is a Riemann integral: γ' is bounded and has at most finitely many discontinuities.

Unless otherwise mentioned, closed contour integrals in the lectures use the anticlockwise orientation.

Source slide 80.

Let $\phi : [a, b] \rightarrow [\alpha, \beta]$ be continuous, strictly increasing, onto, and continuously differentiable. Then $\phi(a) = \alpha$, $\phi(b) = \beta$, and $\sigma = \gamma \circ \phi$ is a path on $[a, b]$. For f continuous on the common trace, $\int_{\sigma} f(z) dz = \int_{\gamma} f(z) dz$. The map ϕ is a change of parameter.

Source slide 81.

If the endpoint of γ_1 is the initial point of γ_2 , a suitable reparametrization allows us to follow first γ_1 , then γ_2 . The resulting path is denoted $\gamma = \gamma_1 + \gamma_2$, and

$$\int_{\gamma} f = \int_{\gamma_1} f + \int_{\gamma_2} f$$

for continuous f on $\gamma_1^* \cup \gamma_2^*$. For paths $\gamma_1, \dots, \gamma_n$ with matching successive endpoints, $\gamma_1 + \cdots + \gamma_n$ is defined similarly.

Source slide 82.

For $\gamma : [0, 1] \rightarrow \mathbb{C}$, the opposite path is $\gamma_1(t) = \gamma(1 - t)$ and $\int_{\gamma} f = -\int_{\gamma_1} f$. Also, for a path on $[\alpha, \beta]$,

$$\begin{aligned} \left| \int_{\gamma} f(z) dz \right| &= \left| \int_{\alpha}^{\beta} f(\gamma(t)) \gamma'(t) dt \right| \\ &\leq \max_{z \in \gamma^*} |f(z)| \int_{\alpha}^{\beta} |\gamma'(t)| dt = \ell(\gamma) \max_{z \in \gamma^*} |f(z)|. \end{aligned}$$

Here $\ell(\gamma) = \int_{\alpha}^{\beta} |\gamma'(t)| dt$ is the length.

Source slide 83.

3.2.4. Remark. For a closed path, $\mathbb{C}_{\infty} \setminus \gamma^*$ is open and is a disjoint union of regions. These are the regions determined by γ . Precisely one contains ∞ , corresponding to the unbounded region. The regions determined in $\mathbb{C}_{\infty}$ and in $\mathbb{C}$ are identical except that the unbounded region in $\mathbb{C}$ does not contain ∞ .

Source slide 84.

3.2.5. Example: Circle. The positively oriented circle of center $\mathbf{a}$ and radius r is $\gamma(t) = \mathbf{a} + re^{it}$, $0 \leq t \leq 2\pi$. Its length is $2\pi r$ and

$$\int_{\gamma} f(z) dz = \int_0^{2\pi} f(\mathbf{a} + re^{it}) ire^{it} dt.$$

Source slide 85.

3.2.6. Example: Segment. The positively oriented interval $[\mathbf{a}, \mathbf{b}]$ is parametrized by $\gamma(t) = \mathbf{a} + (\mathbf{b} - \mathbf{a})t$, $0 \leq t \leq 1$. It has length $|\mathbf{b} - \mathbf{a}|$ and

$$\int_{[\mathbf{a}, \mathbf{b}]} f(z) dz = (\mathbf{b} - \mathbf{a}) \int_0^1 f(\mathbf{a} + (\mathbf{b} - \mathbf{a})t) dt.$$

On an arbitrary $[\alpha, \beta]$, the equivalent parametrization is $\gamma_1(t) = [\mathbf{a}(\beta - t) + \mathbf{b}(t - \alpha)]/(\beta - \alpha)$. The opposite path is $[\mathbf{b}, \mathbf{a}]$.

Source slide 86.

3.2.7. Example: Triangle. For an ordered triple $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$, the triangle $\Delta(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is the smallest convex set containing these points. For continuous f on its boundary, define

$$\int_{\partial\Delta} f = \int_{[\mathbf{a}, \mathbf{b}]} f + \int_{[\mathbf{b}, \mathbf{c}]} f + \int_{[\mathbf{c}, \mathbf{a}]} f.$$

Alternatively, regard $\partial\Delta$ as the concatenation of these paths, and the equality follows from path addition. Cyclic permutation of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ does not change the integral; replacing the order by $\mathbf{a}, \mathbf{c}, \mathbf{b}$ changes its sign.

CHAPTER 4

Index function and Cauchy's theorem

MATH 503, Fall 2025. September 15, 2025. Merged PDF pages 87–113.

4.1 Index and primitives

Source slide 88.

4.1.1. Definition. For a path γ and $a \notin \gamma^*$, write

$$\text{Ind}_\gamma(a) = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z-a}.$$

This is the *index*, or *winding number*, of γ about a .

4.1.2. Theorem. — If γ is closed and $\Omega = \mathbb{C} \setminus \gamma^*$, then Ind_γ is integer-valued, constant on each region of Ω , and zero on the unbounded region.

Source slide 91.

4.1.3. Remark. For a closed curve and a outside its image, the number of times the curve winds around a can be computed from the change of argument of $z-a$ as z travels along the curve. The quantity $\arg(z-a)/(2\pi)$ increases or decreases by one for each complete positive or negative turn. Logarithms will later give a geometric explanation of this interpretation.

Source slide 92.

4.1.4. Theorem. — For the positively oriented circle γ of center a and radius r ,

$$\text{Ind}_\gamma(z) = \begin{cases} 1, & |z-a| < r, \\ 0, & |z-a| > r. \end{cases}$$

Source slide 93.

4.1.5. Definition. A *primitive* of $f : \Omega \rightarrow \mathbb{C}$, where Ω is open, is a holomorphic function F on Ω with $F' = f$.

4.1.6. Theorem. — If f is continuous on Ω and has a primitive F there, then for any path γ from w_1 to w_2 in Ω , $\int_\gamma f(z) dz = F(w_2) - F(w_1)$.

Source slide 95.

4.1.7. Corollary. — If f is continuous on an open set, has a primitive there, and γ is a closed path in that set, then $\int_\gamma f(z) dz = 0$.

4.1.8. Example. The function $f(z) = 1/z$ has no primitive on $\mathbb{C} \setminus \{0\}$. On the unit circle $\mathbb{C}$, parametrized by $z = e^{it}$,

$$\int_{\mathbb{C}} f(z) \, dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}} \, dt = 2\pi i \neq 0.$$

Source slide 96.

4.1.9. Corollary. — If f is holomorphic on a region Ω and $f' = 0$, then f is constant.

4.2 Local Cauchy theorem

Source slide 97.

4.2.1. Theorem (Cauchy–Goursat). — Let $\Omega \subseteq \mathbb{C}$ be open, let Δ be a closed triangle contained in Ω , and let $p \in \Omega$. If f is continuous on Ω and holomorphic on $\Omega \setminus \{p\}$, then $\int_{\partial\Delta} f(z) \, dz = 0$.

4.2.2. Remark. In particular, this holds for every triangular path in Ω when f is holomorphic on Ω . This is Cauchy’s theorem for triangular paths.

Source slide 104.

4.2.3. Theorem (Cauchy’s theorem for convex sets). — Let $\Omega \subseteq \mathbb{C}$ be convex and open, and let $p \in \Omega$. If f is continuous on Ω and holomorphic on $\Omega \setminus \{p\}$, then f has a primitive on Ω , and $\int_{\gamma} f(z) \, dz = 0$ for every closed path γ in Ω .

4.2.4. Corollary. — Every holomorphic function on an open convex set has a primitive; its integral along every closed path in that set is zero.

Source slide 107.

4.2.5. Remark. Cauchy’s theorem also holds for the illustrated regions, although some of them are not convex.

4.2.A. EXERCISE. Explain why Cauchy’s theorem holds in each of these regions.

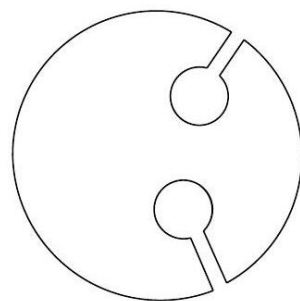
4.3 Worked contour-integral examples

Source slide 108–110.

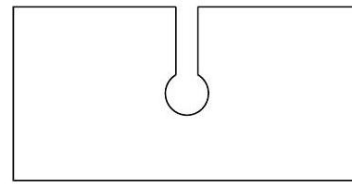
4.3.1. Example: Gaussian Fourier transform. For $\xi \in \mathbb{R}$,

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} \, dx.$$

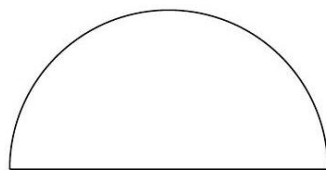
For $\xi = 0$, this is the known Gaussian integral $\int_{\mathbb{R}} e^{-\pi x^2} \, dx = 1$. For $\xi > 0$, use the entire function $f(z) = e^{-\pi z^2}$ and the positively oriented rectangle with vertices $R, R + i\xi, -R + i\xi, -R$. Cauchy’s theorem gives $\int_{\gamma_R} f(z) \, dz = 0$. The bottom side



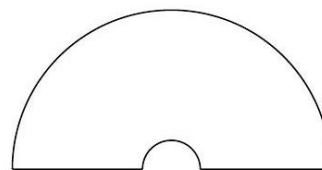
The multiple keyhole



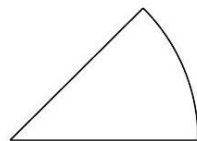
Rectangular keyhole



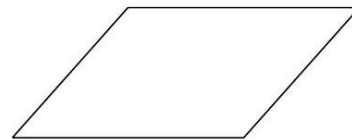
Semicircle



Indented semicircle



Sector



Parallelogram

FIGURE 1. The original multiple keyhole, rectangular keyhole, semicircle, indented semicircle, sector, and parallelogram (slide 107).

γ_R given here

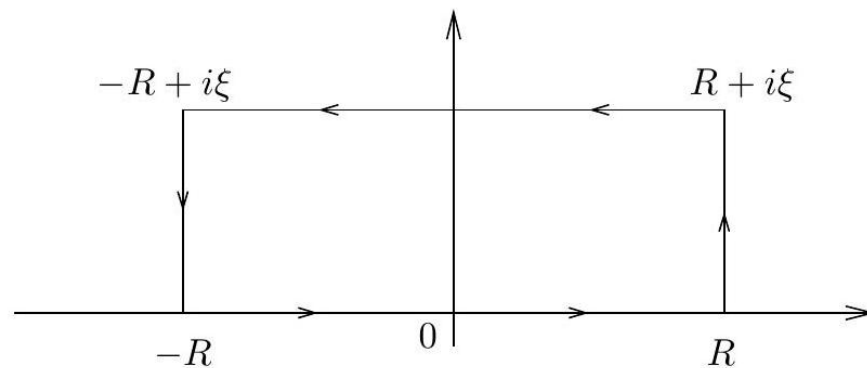


FIGURE 2. The rectangle γ_R used for the Gaussian transform.

contributes $\int_{-R}^R e^{-\pi x^2} dx \rightarrow 1$. On the right side,

$$I(R) = \int_0^\xi f(R + iy)i dy = i \int_0^\xi e^{-\pi(R^2 + 2iRy - y^2)} dy.$$

For fixed ξ , $|I(R)| \leq Ce^{-\pi R^2} \rightarrow 0$. The left side has the same vanishing limit. The top contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2} dx = -e^{\pi\xi^2} \int_{-R}^R e^{-\pi x^2} e^{-2\pi i x \xi} dx.$$

Thus the limiting identity is $0 = 1 - e^{\pi\xi^2} \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x \xi} dx$. For $\xi < 0$, use the corresponding rectangle in the lower half-plane.

Source slide 111–113.

4.3.2. Example. Another classical integral is $\int_0^\infty (1 - \cos x)/x^2 dx = \pi/2$. Take $f(z) = (1 - e^{iz})/z^2$ and an indented semicircle in the upper half-plane. Denote

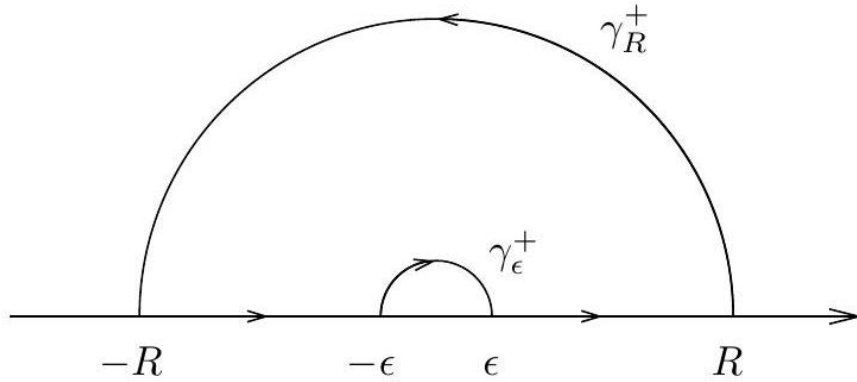


FIGURE 3. The small negatively oriented and large positively oriented semicircles in the original example.

the small and large arcs by γ_ϵ^+ and γ_R^+ ; their orientations are negative and positive, respectively. Cauchy's theorem yields

$$\int_{-R}^{-\epsilon} f(x) dx + \int_{\gamma_\epsilon^+} f(z) dz + \int_\epsilon^R f(x) dx + \int_{\gamma_R^+} f(z) dz = 0.$$

On the large arc, $|f(z)| \leq 2/|z|^2$, so its integral tends to zero as $R \rightarrow \infty$. Consequently

$$\int_{|x| \geq \epsilon} \frac{1 - e^{ix}}{x^2} dx = - \int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2} dz.$$

Near zero, $f(z) = -i/z + E(z)$, where E is bounded. Parametrizing the small arc by $z = \epsilon e^{i\theta}$, with θ decreasing from π to 0 , gives

$$\lim_{\epsilon \rightarrow 0} \int_{\gamma_\epsilon^+} f(z) dz = \int_\pi^0 (-i) i d\theta = -\pi.$$

Taking real parts gives $\int_{\mathbb{R}} (1 - \cos x)/x^2 \, dx = \pi$. The integrand is even, giving the stated half-line integral.

CHAPTER 5

Cauchy's integral formula and Laurent series

MATH 503, Fall 2025. September 18, 2025. Merged PDF pages 114–139.

5.1 Integral formula and its consequences

Source slide 115.

5.1.1. Theorem (Recall: Cauchy's theorem for convex sets). — If Ω is open and convex, $p \in \Omega$, and f is continuous on Ω and holomorphic on $\Omega \setminus \{p\}$, then f has a primitive on Ω and $\int_{\gamma} f(z) dz = 0$ for every closed path γ in Ω .

5.1.2. Theorem (Cauchy's integral formula for convex sets). — Let γ be a closed path in an open convex set $\Omega \subseteq \mathbb{C}$ and $f \in H(\Omega)$. For every $a \in \Omega \setminus \gamma^*$,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz = \text{Ind}_{\gamma}(a) f(a).$$

Source slide 117.

5.1.3. Corollary. — If $f \in H(\Omega)$, then $f^{(n)} \in H(\Omega)$ for every $n \in \mathbb{N}$. If a positively oriented circle C and its interior lie in Ω , then, for z in that interior,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Thus holomorphic functions have infinitely many complex derivatives.

Source slide 120.

5.1.4. Corollary (Cauchy's inequality). — If f is holomorphic on an open set containing $\overline{D(z_0, R)}$, and $C = \partial D(z_0, R)$, then

$$|f^{(n)}(z_0)| \leq \frac{n!}{R^n} \|f\|_{L^\infty(C)}, \quad \|f\|_{L^\infty(C)} = \sup_{z \in C} |f(z)|.$$

Source slide 121.

5.1.5. Theorem. — If $f \in H(\Omega)$ and D is a disc centered at z_0 with $\overline{D} \subset \Omega$, then, for every $z \in D$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{f^{(n)}(z_0)}{n!} \quad (n \geq 0).$$

Since power series are indefinitely complex differentiable, this provides another route to the infinite differentiability of holomorphic functions.

Source slide 124.

5.1.6. Remark. The expansion about z_0 converges in any disc, however large, whose closure lies in Ω . In particular, an entire function has an expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converging on all of $\mathbb{C}$.

Source slide 125.

5.1.7. Theorem (Concluding equivalences). — For an open set $\Omega \subseteq \mathbb{C}$, the following are equivalent:

- (a) $f \in H(\Omega)$.
- (b) $f^{(n)} \in H(\Omega)$ for every $n \geq 0$.
- (c) On every disc D centered at z_0 with $\overline{D} \subset \Omega$, $f(z) = \sum_{n \geq 0} a_n (z - z_0)^n$, with $a_n = f^{(n)}(z_0)/n!$.

5.2 Laurent series

Source slide 126.

5.2.1. Definition. For $0 \leq s_1 < s_2 \leq +\infty$,

$$A(z_0, s_1, s_2) = \{z : s_1 < |z - z_0| < s_2\}$$

is an open annulus. For finite radii, the closed annulus is $\overline{A}(z_0, s_1, s_2) = \{z : s_1 \leq |z - z_0| \leq s_2\}$. In the usual positive finite-radius case,

$$\begin{aligned} A(z_0, s_1, s_2) &= D(z_0, s_2) \setminus \overline{D}(z_0, s_1), \\ \overline{A}(z_0, s_1, s_2) &= \overline{D}(z_0, s_2) \setminus D(z_0, s_1), \\ \overline{A}(z_0, s_1, s_2) &= A(z_0, s_1, s_2) \cup C(z_0, s_1) \cup C(z_0, s_2). \end{aligned}$$

Editorial clarification. At radius zero, use the distance definitions; the inner boundary degenerates to the center. An infinite radius does not give a finite boundary circle.

Source slide 127.

5.2.2. Definition. A *Laurent series* about z_0 is $\sum_{n \in \mathbb{Z}} a_n (z - z_0)^n$. Its convergence means that both $\sum_{n \geq 0} a_n (z - z_0)^n$ and $\sum_{n < 0} a_n (z - z_0)^n$ converge; its value is their sum. It diverges if either tail diverges.

Source slide 128.

5.2.3. Theorem. — For a Laurent series, put

$$r = \limsup_{n \rightarrow \infty} |a_{-n}|^{1/n}, \quad R = \left(\limsup_{n \rightarrow \infty} |a_n|^{1/n} \right)^{-1}.$$

If $r < R$, the series converges absolutely on $A(z_0, r, R)$, uniformly on its compact subsets, and defines a holomorphic function there.

Source slide 131.

5.2.4. Theorem (Cauchy's formula for annuli). — Let $0 < r_1 < r_2 < \infty$, and let f be holomorphic on an open neighborhood of $\overline{A}(z_0, r_1, r_2)$. If γ_1, γ_2 are its positively oriented inner and outer circles, then

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w-z} dw - \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w-z} dw \quad (z \in A(z_0, r_1, r_2)).$$

5.2.A. EXERCISE (Question retained from the omitted argument). Explain in what sense Cauchy's integral formula is available on the annular region $A(z_0, s_1, s_2)$ with $0 < s_1 < r_1 < r_2 < s_2$.

Editorial clarification. Use the annular boundary formula, for example by cutting along a radial segment. The assertion that every holomorphic function on an annulus has a primitive would be false.

Source slide 133.

5.2.5. Theorem (Laurent representation). — If $f \in H(A(z_0, s_1, s_2))$, there is a unique sequence $(a_n)_{n \in \mathbb{Z}}$ such that

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

For $s_1 < r < s_2$ and $n \in \mathbb{Z}$,

$$a_n = \frac{1}{2\pi i} \int_{C(z_0, r)} \frac{f(w)}{(w - z_0)^{n+1}} dw.$$

The expansion converges absolutely on the annulus and uniformly on its compact subsets.

Source slide 138.

5.2.B. EXERCISE (Question retained from the omitted argument). Suppose $f(z) = \sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$ for every $z \in A(z_0, s_1, s_2)$. Why must this series converge uniformly on compact subsets of that annulus?

CHAPTER 6

Further applications and singularities

MATH 503, Fall 2025. September 22, 2025. Merged PDF pages 140–165.

6.1 Further applications of Cauchy's formula

Source slide 141.

6.1.1. Corollary (Cauchy's inequality, recalled). — If f is holomorphic on an open neighborhood of $\overline{D}(z_0, R)$ and $C = \partial D(z_0, R)$, then

$$|f^{(n)}(z_0)| \leq n! R^{-n} \|f\|_{L^\infty(C)}, \quad \|f\|_{L^\infty(C)} = \sup_{z \in C} |f(z)|.$$

Source slide 142.

6.1.2. Theorem (Liouville). — Every bounded entire function is constant.

6.1.3. Theorem (Fundamental theorem of algebra). — Every nonconstant polynomial $P(z) = a_n z^n + \cdots + a_1 z + a_0$ with complex coefficients has a root in $\mathbb{C}$. If its degree is $n \geq 1$, it has exactly n roots, counted with multiplicity.

Source slide 144.

6.1.4. Theorem (Morera: converse to Cauchy's theorem). — If f is continuous on an open set $\Omega \subseteq \mathbb{C}$ and $\int_{\partial \Delta} f(z) dz = 0$ for every closed triangle $\Delta \subset \Omega$, then $f \in H(\Omega)$.

Source slide 146.

6.1.5. Definition. The sequence (f_j) converges to f uniformly on compact subsets of Ω if for every compact $K \subseteq \Omega$ and $\varepsilon > 0$ there exists $N = N(K, \varepsilon)$ such that $\sup_{z \in K} |f_j(z) - f(z)| < \varepsilon$ whenever $j > N$.

6.1.6. Theorem. — If $(f_j) \subset H(\Omega)$ converges uniformly on compact subsets to f , then $f \in H(\Omega)$ and $f'_j \rightarrow f'$ uniformly on compact subsets.

6.2 Zeros and analytic continuation

Source slide 148.

6.2.1. Theorem. — Let Ω be a region, $f \in H(\Omega)$, and $Z(f) = \{a \in \Omega : f(a) = 0\}$. Either $Z(f) = \Omega$, or $Z(f)$ has no limit point in Ω . In the latter case, for each $a \in Z(f)$ there is a unique positive integer $m = m(a)$ with

$$f(z) = (z - a)^m g(z) \quad (z \in \Omega), \quad g \in H(\Omega), \quad g(a) \neq 0.$$

Moreover, $Z(f)$ is at most countable. The integer m is the order of the zero at a .

Source slide 153.

6.2.2. Theorem (Identity theorem). — If f, g are holomorphic on a region Ω and agree on a set having a limit point in Ω , then $f = g$ throughout Ω .

6.2.3. Definition. If f, F are analytic on regions Ω, Ω' with $\Omega \subseteq \Omega'$ and $F = f$ on Ω , then F is an *analytic continuation* of f to Ω' . The identity theorem guarantees uniqueness of this continuation.

Source slide 154–155.

Let Ω be open and symmetric about the real axis: $z \in \Omega$ if and only if $\bar{z} \in \Omega$. Let Ω^+ and Ω^- be its upper and lower half-plane parts, and set $I = \Omega \cap \mathbb{R}$. Thus $\Omega = \Omega^+ \cup I \cup \Omega^-$. The set I is the relative interior of the real-axis portion of their common boundary.

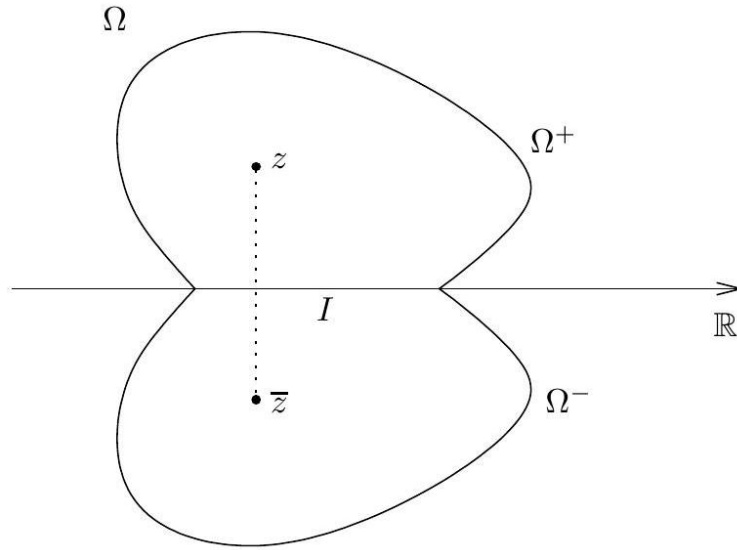


FIGURE 1. The symmetric region from slide 154.

6.2.4. Theorem (Symmetry principle). — If f^+, f^- are holomorphic on Ω^+, Ω^- , extend continuously to I , and $f^+(x) = f^-(x)$ there, then

$$f(z) = \begin{cases} f^+(z), & z \in \Omega^+, \\ f^+(z) = f^-(z), & z \in I, \\ f^-(z), & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω .

Source slide 158.

6.2.5. Theorem (Schwarz reflection). — If f is holomorphic on Ω^+ , extends continuously to $\mathbb{I}$, and is real-valued on $\mathbb{I}$, then a holomorphic function F on Ω exists with $F = f$ on Ω^+ .

6.3 Classification of singularities

Source slide 159.

6.3.1. Definition. If $a \in \Omega$ and $f \in H(\Omega \setminus \{a\})$, then a is an *isolated singularity*. It is *removable* if f can be defined at a so as to be holomorphic on Ω .

6.3.2. Theorem (Riemann's removable-singularity theorem). — If $f \in H(\Omega \setminus \{a\})$ is bounded on a punctured disc $D'(a, r)$, then its singularity at a is removable.

Source slide 161.

6.3.3. Theorem. — For $a \in \Omega$ and $f \in H(\Omega \setminus \{a\})$, exactly one of the following occurs:

- (a) The singularity is removable.
- (b) There are $c_1, \dots, c_m \in \mathbb{C}$, with $m \geq 1$ and $c_m \neq 0$, such that $f(z) - \sum_{k=1}^m c_k(z - a)^{-k}$ has a removable singularity.
- (c) For every $r > 0$ with $D(a, r) \subseteq \Omega$, $f(D'(a, r))$ is dense in $\mathbb{C}$.

The last conclusion is the Casorati–Weierstrass theorem.

Source slide 162.

6.3.4. Remark. In the second case, f has a *pole of order m* . The polynomial in $(z - a)^{-1}$, $\sum_{k=1}^m c_k(z - a)^{-k}$, is its *principal part*; then $|f(z)| \rightarrow \infty$ as $z \rightarrow a$. In the third case, the singularity is *essential*. Equivalently, for every $w \in \mathbb{C}$ there is a sequence $z_n \rightarrow a$ with $f(z_n) \rightarrow w$.

Source slide 165.

6.3.5. Remark. For the Laurent expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$:

- The singularity is removable exactly when $a_n = 0$ for every $n < 0$.
- It is a pole of order m when m is the largest positive integer with $a_{-m} \neq 0$. A pole of order one is *simple*.
- It is essential when infinitely many negative-index coefficients are nonzero.
- The behavior at infinity is studied using $g(z) = f(1/z)$ near zero. The point ∞ is an isolated singularity if f is analytic for $|z| > r$ for some r ; its type is the type of g at zero.

CHAPTER 7

Maximum principle, open mappings, and logarithms

MATH 503, Fall 2025. September 25, 2025. Merged PDF pages 166–195.

7.1 Maximum modulus and open mappings

Source slide 167.

7.1.1. Theorem (Orthogonality relation). — If $f(z) = \sum_{n \geq 0} c_n(z - a)^n$ on $D(a, R)$ and $0 < r < R$, then

$$\sum_{n \geq 0} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(a + re^{i\theta})|^2 d\theta.$$

Source slide 169.

7.1.2. Definition. The modulus $|f|$ has a local maximum at $a \in \Omega$ if some $\delta > 0$ satisfies $D(a, \delta) \subseteq \Omega$ and $|f(a)| \geq |f(z)|$ throughout this disc. Having no local maximum means having none at any point of Ω . A local minimum is defined with the inequality reversed.

7.1.3. Theorem (Maximum modulus principle). — If Ω is a region and $f \in H(\Omega)$, then $|f|$ has no local maximum unless f is constant. If additionally $\overline{\Omega}$ is compact and f is continuous on $\overline{\Omega}$, then $\sup_{\Omega} |f| \leq \sup_{\partial\Omega} |f|$.

Source slide 172.

7.1.A. EXERCISE (Corollary to prove). Let Ω be a region, $f \in H(\Omega)$, and $\overline{D(a, r)} \subset \Omega$. Prove that

$$|f(a)| \leq \max_{-\pi \leq \theta \leq \pi} |f(a + re^{i\theta})|,$$

with equality if and only if f is constant on Ω . Deduce the absence of local maxima for nonconstant $|f|$. If f has no zero in $D(a, r)$, prove also that

$$|f(a)| \geq \min_{-\pi \leq \theta \leq \pi} |f(a + re^{i\theta})|.$$

Source slide 173.

7.1.4. Lemma. — If $f \in H(\Omega)$, then

$$g(z, w) = \begin{cases} (f(z) - f(w))/(z - w), & z \neq w, \\ f'(z), & z = w \end{cases}$$

is continuous on $\Omega \times \Omega$.

Source slide 175.

7.1.5. Theorem. — If $\varphi \in H(\Omega)$ and $\varphi'(z_0) \neq 0$, there is a neighborhood $V \subseteq \Omega$ of z_0 on which φ is one-to-one, $W = \varphi(V)$ is open, and the inverse $\psi : W \rightarrow V$, defined by $\psi(\varphi(z)) = z$, is holomorphic.

Source slide 178.

7.1.6. Definition. For $m \in \mathbb{N}$, write $\pi_m(z) = z^m$.

Every $w = re^{i\theta} \neq 0$ has exactly m distinct preimages:

$$\pi_m(z) = w \iff z = r^{1/m} e^{i(\theta + 2k\pi)/m}, \quad k = 1, \dots, m.$$

Each π_m is open: away from zero use the preceding theorem, and at zero use $\pi_m(D(0, r)) = D(0, r^m)$. Compositions of open maps are open. In particular, $\pi_m \circ \varphi$ is open when φ' has no zero.

Source slide 179.

7.1.7. Theorem (Local form and open mapping theorem). — Let f be nonconstant and holomorphic on a region Ω , $z_0 \in \Omega$, and $w_0 = f(z_0)$. If m is the order of the zero of $f - w_0$ at z_0 , there are a neighborhood V of z_0 and $\varphi \in H(V)$ such that

$$f(z) = w_0 + \varphi(z)^m.$$

Moreover, φ' has no zero on V , and φ maps V invertibly onto a disc $D(0, r)$.

7.1.8. Remark. Thus $f - w_0 = \pi_m \circ \varphi$ locally. The map $f : V \setminus \{z_0\} \rightarrow D'(w_0, r^m)$ is exactly m -to-one. Every $w_0 \in f(\Omega)$ is an interior point, so $f(\Omega)$ is open.

Source slide 181.

7.1.9. Theorem (Inverse mapping theorem). — If $f \in H(\Omega)$ is one-to-one on a region Ω , then $f'(z) \neq 0$ everywhere and f^{-1} is holomorphic.

7.1.10. Remark. The converse is false: $f(z) = e^z$ has nonvanishing derivative everywhere but is not one-to-one on $\mathbb{C}$.

7.2 Logarithms and continuous arguments

Source slide 182.

The exponential restricted to $S_\alpha = \{x + iy : \alpha \leq y < \alpha + 2\pi\}$ is a bijection onto $\mathbb{C} \setminus \{0\}$.

7.2.1. Definition. Its inverse is $\log_\alpha$, and $\arg_\alpha = \Im \log_\alpha$. Thus $\log_\alpha(e^z) = z$ on S_α and $e^{\log_\alpha z} = z$ for $z \neq 0$. The principal branches used here are $\text{Log} = \log_{-\pi}$ and $\text{Arg} = \arg_{-\pi}$.

Source slide 183.

- 7.2.2. Theorem.** — (a) For $z \neq 0$, $\log_\alpha z = \log|z| + i \arg_\alpha z$, where $\arg_\alpha z$ is the unique argument in $[\alpha, \alpha + 2\pi)$; equivalently $z/|z| = e^{i \arg_\alpha z}$.
 (b) Set $R_\alpha = \{re^{i\alpha} : r \geq 0\}$. The two functions are continuous on $\mathbb{C} \setminus R_\alpha$ and discontinuous on the nonzero part of the slit.
 (c) $\log_\alpha$ is analytic on $\mathbb{C} \setminus R_\alpha$ and $(\log_\alpha)'(z) = 1/z$.

Editorial clarification. Neither function is defined at zero. The half-closed strip is not an open domain of holomorphy; the exponential is the restriction of an entire function.

Source slide 185.

7.2.3. Theorem (Recalled inverse criterion). — Let g be analytic on Ω_1 and f continuous on the open set Ω . If $f(\Omega) \subseteq \Omega_1$, g' is nowhere zero, and $g(f(z)) = z$, then f is analytic and $f' = 1/(g' \circ f)$. The identity also makes f one-to-one.

Source slide 186.

7.2.4. Definition. Let $S \subseteq \mathbb{C}$, or more generally let S be a metric space, and let $f : S \rightarrow \mathbb{C} \setminus \{0\}$ be continuous. A continuous logarithm is a continuous $g : S \rightarrow \mathbb{C}$ with $f = e^g$. A continuous argument is a continuous $\theta : S \rightarrow \mathbb{R}$ with $f = |f|e^{i\theta}$.

- 7.2.5. Example.** (a) On $S = [0, 2\pi]$, $f(s) = e^{is}$ has continuous arguments $\theta(s) = s + 2k\pi$ for any fixed integer k .
 (b) If $f : S \rightarrow \mathbb{C} \setminus R_\alpha$ is continuous, $\theta = \arg_\alpha \circ f$ is a continuous argument.
 (c) On $S = \{z : |z| = 1\}$, $f(z) = z$ has no continuous argument.

Source slide 187.

7.2.6. Theorem. — For continuous nonvanishing $f : S \rightarrow \mathbb{C}$:

- (a) If g is a continuous logarithm, $\Im g$ is a continuous argument.
 (b) If θ is a continuous argument, $\log|f| + i\theta$ is a continuous logarithm. Thus either exists if and only if the other does.
 (c) On connected S , any two continuous logarithms differ by $2\pi ik$ for one fixed $k \in \mathbb{Z}$; any two continuous arguments differ by $2\pi l$ for one fixed $l \in \mathbb{Z}$.
 (d) If S is connected and $s, t \in S$, any continuous logarithm g and argument θ satisfy

$$g(s) - g(t) = \log|f(s)| - \log|f(t)| + i(\theta(s) - \theta(t)).$$

Source slide 189.

7.2.7. Theorem. — Every continuous curve $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{0\}$ has a continuous argument and hence a continuous logarithm.

Source slide 190.

7.2.8. Theorem. — For a closed curve $\gamma : [a, b] \rightarrow \mathbb{C}$, $z_0 \notin \gamma^*$, and any continuous argument θ of $\gamma - z_0$, $\theta(b) - \theta(a)$ is an integer multiple of 2π and is independent of the choice of θ .

Source slide 191.

7.2.9. Definition. The winding number is $W(\gamma, z_0) = (\theta(b) - \theta(a))/(2\pi)$.

7.2.10. Remark. It is well-defined by the preceding theorem. Intuitively it is the net number of revolutions about z_0 as the parameter runs from a to b . It is translation-invariant: $W(\gamma, z_0) = W(\gamma + w, z_0 + w)$ for every $w \in \mathbb{C}$.

Source slide 192.

7.2.11. Theorem. — For a closed path,

$$W(\gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - z_0} = \text{Ind}_{\gamma}(z_0).$$

More generally, if f is analytic on a neighborhood of γ^* and $z_0 \notin (f \circ \gamma)^*$, then

$$W(f \circ \gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z) - z_0} dz.$$

Global Cauchy's theorem

MATH 503, Fall 2025. October 2, 2025. Merged PDF pages 196–225.

8.1 Chains and cycles

Source slide 197–198.

For paths $\gamma_1, \dots, \gamma_n$, put $K = \bigcup_i \gamma_i^*$. On the vector space $C(K)$ of continuous functions on K , each path induces the linear functional $\tilde{\gamma}_i(f) = \int_{\gamma_i} f(z) dz$. Define $\tilde{\Gamma} = \sum_i \tilde{\gamma}_i$; explicitly $\tilde{\Gamma}(f) = \sum_i \tilde{\gamma}_i(f)$. This motivates the formal sum $\Gamma = \gamma_1 + \dots + \gamma_n$ and the definition

$$\int_{\Gamma} f(z) dz = \tilde{\Gamma}(f) = \sum_{i=1}^n \int_{\gamma_i} f(z) dz.$$

8.1.1. Definition. Such a formal sum is a *chain* in Ω if each $\gamma_i^* \subset \Omega$. It is a *cycle* if it has a representation in which every γ_i is closed. Equivalently, the initial and terminal endpoints in any representation can be paired. For the displayed representation, write $\Gamma^* = \bigcup_i \gamma_i^*$.

The sum is meaningful as addition of functionals, not as pointwise addition of paths. The endpoint criterion follows from a combinatorial argument.

Source slide 199–200.

For a cycle and $\alpha \notin \Gamma^*$, set

$$\text{Ind}_{\Gamma}(\alpha) = \frac{1}{2\pi i} \int_{\Gamma} \frac{dz}{z - \alpha} = \sum_i \text{Ind}_{\gamma_i}(\alpha).$$

Replacing each path by its opposite gives $-\Gamma$; then $\int_{-\Gamma} f = -\int_{\Gamma} f$, and $\text{Ind}_{-\Gamma}(\alpha) = -\text{Ind}_{\Gamma}(\alpha)$. Chains can be added or subtracted through their functionals. For integers k_1, k_2 ,

$$\Gamma = k_1 \Gamma_1 + k_2 \Gamma_2 \implies \int_{\Gamma} f = k_1 \int_{\Gamma_1} f + k_2 \int_{\Gamma_2} f$$

for every f continuous on $\Gamma_1^* \cup \Gamma_2^*$. Chains can have many representations: equality of $\sum_i \gamma_i$ and $\sum_j \delta_j$ means equality of $\sum_i \int_{\gamma_i} f$ and $\sum_j \int_{\delta_j} f$ for every continuous f on the union of all the path images. In particular, a cycle can be represented by paths that are not closed.

8.2 Global integral theorem and formula

Source slide 201.

8.2.1. Theorem. — Let $f \in H(\Omega)$ for an arbitrary open $\Omega \subseteq \mathbb{C}$. If a cycle Γ in Ω satisfies $\text{Ind}_\Gamma(\alpha) = 0$ for every $\alpha \notin \Omega$, then

$$f(z) \text{Ind}_\Gamma(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(w)}{w-z} dw \quad (z \in \Omega \setminus \Gamma^*), \quad \int_\Gamma f(z) dz = 0.$$

If cycles Γ_0, Γ_1 have equal indices at every $\alpha \notin \Omega$, then $\int_{\Gamma_0} f = \int_{\Gamma_1} f$.

Source slide 206.

8.2.2. Corollary. — For a cycle Γ in an open set Ω , the following are equivalent:

- (a) The preceding Cauchy integral formula holds for every $f \in H(\Omega)$.
- (b) $\int_\Gamma f(z) dz = 0$ for every $f \in H(\Omega)$.
- (c) $\text{Ind}_\Gamma(\alpha) = 0$ for every $\alpha \in \mathbb{C} \setminus \Omega$.

Editorial clarification. The universal quantifier on f is essential. The slide introduces one f before the equivalence, but the assertion is false for a single fixed function, for example $f = 0$.

Source slide 208–210.

8.2.3. Remark. In a convex region Ω , apply the convex Cauchy theorem to $(z-\alpha)^{-1}$, $\alpha \notin \Omega$, to obtain $\text{Ind}_\gamma(\alpha) = 0$. Thus every cycle there satisfies the global hypothesis, which generalizes the earlier convex results. An efficient way to compute index is the winding number:

$$W(\gamma, z_0) = \frac{\theta_{z_0}(b) - \theta_{z_0}(a)}{2\pi} = \frac{1}{2\pi i} \int_\gamma \frac{dw}{w - z_0} = \text{Ind}_\gamma(z_0).$$

It counts net turns. Each anticlockwise loop increases the argument divided by 2π by one, and each clockwise loop decreases it by one. The last part of the theorem specifies when one integration cycle can be replaced by another without changing the integral.

Source slide 211.

8.2.4. Example. For the illustrated path in $\mathbb{U}$ and $f \in H(\mathbb{U})$, $\text{Ind}_\gamma(z_1) = \text{Ind}_\gamma(z_2) = \text{Ind}_\gamma(z_3) = 1$. Taking small anticlockwise circles about these points gives $\int_\gamma f = \int_{\gamma_1} f + \int_{\gamma_2} f + \int_{\gamma_3} f$.

Source slide 212.

8.2.5. Example. In $\mathbb{C} \setminus \{z_1, z_2, z_3\}$, the second illustrated path has indices 1, 2, 2. For positively oriented small circles and holomorphic f , $\int_\gamma f = \int_{\gamma_1} f + 2 \int_{\gamma_2} f + 2 \int_{\gamma_3} f$.

Source slide 213.

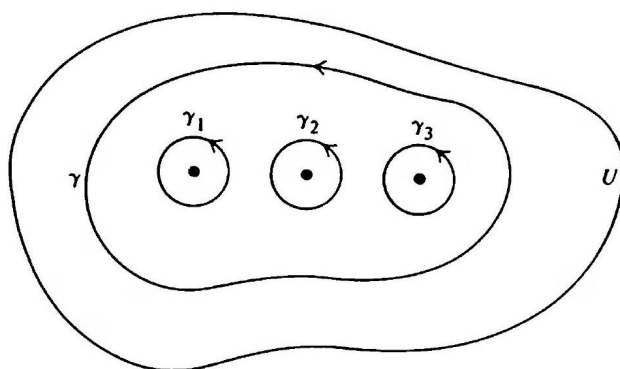


FIGURE 1. First cycle-deformation example.

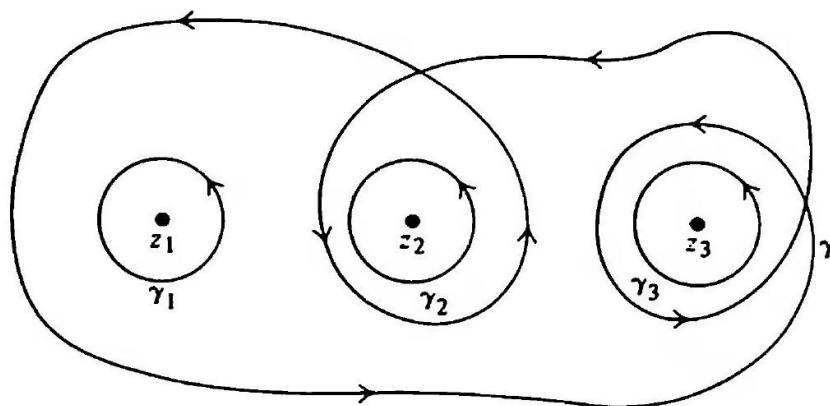


FIGURE 2. Second cycle-deformation example.

8.2.6. Example. In $\mathbb{C} \setminus \{z_1, z_2, z_3, z_4\}$, the third path has indices $-1, -2, -1, -2$. Using positively oriented small circles,

$$\int_{\gamma} f = - \int_{\gamma_1} f - 2 \int_{\gamma_2} f - \int_{\gamma_3} f - 2 \int_{\gamma_4} f.$$

Editorial clarification. Slide 213 calls the small circles clockwise, but its arrows and negative coefficients use anticlockwise circles. With clockwise small circles, all four coefficients in the formula become positive.

Source slide 214.

Recall the annular formula used for Laurent expansions.

8.2.7. Theorem. — For f analytic near $\overline{A}(z_0, r_1, r_2)$, and positively oriented inner and outer circles γ_1, γ_2 ,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w-z} dw - \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w-z} dw \quad (r_1 < |z-z_0| < r_2).$$

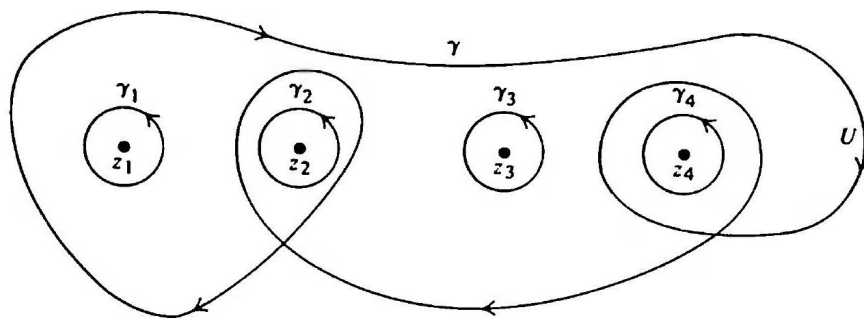


FIGURE 3. Third cycle-deformation example.

8.3 Homotopy and simple connectivity

Source slide 215.

8.3.1. Definition. Closed curves $\gamma_0, \gamma_1 : [0, 1] \rightarrow X$ are X -homotopic if a continuous $H : [0, 1]^2 \rightarrow X$ satisfies

$$H(s, 0) = \gamma_0(s), \quad H(s, 1) = \gamma_1(s), \quad H(0, t) = H(1, t).$$

The curves $\gamma_t(s) = H(s, t)$ form a one-parameter family continuously deforming one curve into the other inside X . A curve homotopic to a constant curve is *null-homotopic*.

Source slide 216.

8.3.2. Definition. A connected space X is *simply connected* if every closed curve in X is null-homotopic.

8.3.3. Example. Every convex region is simply connected: fix $z_1 \in \Omega$ and set $H(s, t) = (1 - t)\gamma_0(s) + tz_1$.

8.3.4. Lemma. — For closed paths $\gamma_0, \gamma_1 : [0, 1] \rightarrow \mathbb{C}$, if $|\gamma_1(s) - \gamma_0(s)| < |\alpha - \gamma_0(s)|$ for all s , then $\text{Ind}_{\gamma_1}(\alpha) = \text{Ind}_{\gamma_0}(\alpha)$.

Source slide 220.

8.3.5. Theorem. — Ω -homotopic closed paths in a region Ω have equal indices at every point outside Ω .

Source slide 224.

8.3.6. Theorem. — In a simply connected region Ω , every closed path Γ has $\text{Ind}_{\Gamma}(\alpha) = 0$ for $\alpha \notin \Omega$. For continuous closed curves, the same assertion uses winding number.

Source slide 225.

8.3.7. Remark. The global Cauchy theorem therefore applies in simply connected regions. For Ω -homotopic closed paths, equality of their outside indices combined with the global theorem gives $\int_{\Gamma_0} f(z) \, dz = \int_{\Gamma_1} f(z) \, dz$ for $f \in H(\Omega)$.

CHAPTER 9

Analytic branches of the logarithm

MATH 503, Fall 2025. October 6, 2025. Merged PDF pages 226–248.

9.1 Branches and arguments, recalled

Source slide 227–228.

The exponential on $S_\alpha = \{x + iy : \alpha \leq y < \alpha + 2\pi\}$ is a bijection onto $\mathbb{C} \setminus \{0\}$. Its inverse is $\log_\alpha$, with $\arg_\alpha = \Im \log_\alpha$. Thus $\log_\alpha(e^z) = z$ on this strip and $e^{\log_\alpha z} = z$ for $z \neq 0$. The principal branches are $\text{Log} = \log_{-\pi}$ and $\text{Arg} = \arg_{-\pi}$.

9.1.1. Theorem. — For $z \neq 0$, $\log_\alpha z = \log |z| + i \arg_\alpha z$, where $\arg_\alpha z \in [\alpha, \alpha + 2\pi)$ is uniquely determined by $z/|z| = e^{i \arg_\alpha z}$. On the slit plane $\mathbb{C} \setminus \mathbb{R}_\alpha$, with $\mathbb{R}_\alpha = \{re^{i\alpha} : r \geq 0\}$, both functions are continuous, and $\log_\alpha$ is analytic with derivative $1/z$. They are discontinuous at each nonzero point of the slit.

Source slide 230.

9.1.2. Theorem (Recalled inverse criterion). — If g is analytic on Ω_1 , $g' \neq 0$, f is continuous on open Ω , $f(\Omega) \subseteq \Omega_1$, and $g(f(z)) = z$, then f is analytic, one-to-one, and $f' = 1/(g' \circ f)$.

9.2 Analytic logarithms and powers

Source slide 231.

9.2.1. Definition. For analytic f on Ω , an *analytic logarithm* is an analytic function g on Ω such that $e^{g(z)} = f(z)$.

9.2.2. Remark. The exponential is not injective on $\mathbb{C}$, so logarithms require a branch. Also, f must be nonvanishing since an exponential is never zero. On $A = \{z : 1/2 < |z| < 2\}$ there is no analytic g with $e^{g(z)} = z$: differentiation would give $g'(z)e^{g(z)} = 1$, hence $g' = 1/z$, but $1/z$ has no primitive on this annulus.

Editorial clarification. The slide concludes that the domain cannot have holes. More precisely, holes can obstruct a logarithm of a particular function; some functions, such as $f = 1$, have logarithms on every domain.

Source slide 232.

9.2.3. Theorem. — If Ω is a simply connected region with $1 \in \Omega$ and $0 \notin \Omega$, there is a branch $F = \log_{\Omega}$ which is holomorphic, satisfies $e^{F(z)} = z$, and agrees with $\log r$ for positive real r sufficiently near 1. Thus this branch extends the standard real logarithm locally near 1.

Source slide 235–236.

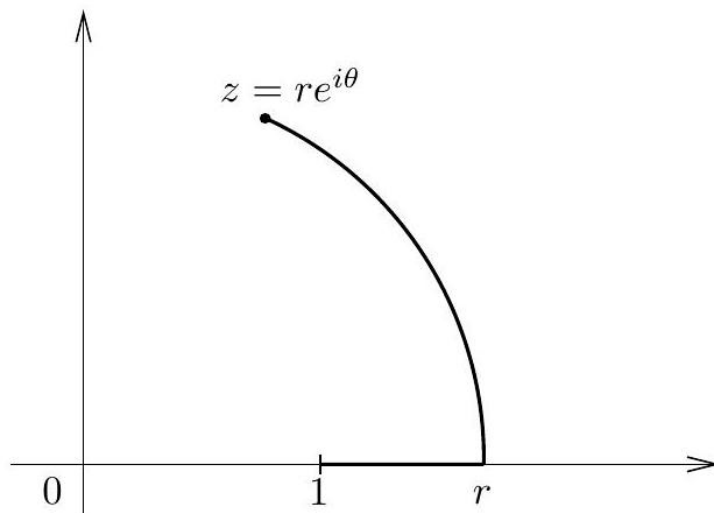


FIGURE 1. Principal branch of the logarithm on the slit plane (source slide 235).

9.2.4. Example. On $\Omega = \mathbb{C} \setminus (-\infty, 0]$, write $z = re^{i\theta}$ with $|\theta| < \pi$. The principal branch is $\log z = \log r + i\theta$. The integration path consists of the real segment from 1 to r followed by the circular arc η from r to z :

$$\log z = \int_1^r \frac{dx}{x} + \int_{\eta} \frac{dw}{w} = \log r + \int_0^{\theta} \frac{ire^{it}}{re^{it}} dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$. For $z_1 = z_2 = e^{2\pi i/3}$, their logarithms are $2\pi i/3$, whereas $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$.

Source slide 237.

For the principal branch, the Taylor expansion is

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} z^n, \quad |z| < 1.$$

Both sides have derivative $1/(1+z)$ and value zero at zero.

Source slide 238.

On such a simply connected domain, choose $\log 1 = 0$ and define, for any $\alpha \in \mathbb{C}$, $z^{\alpha} = e^{\alpha \log z}$. Then $1^{\alpha} = 1$. For $\alpha = 1/n$,

$$(z^{1/n})^n = \prod_{k=1}^n e^{(\log z)/n} = e^{\sum_{k=1}^n (\log z)/n} = e^{\log z} = z.$$

Source slide 239.

9.2.5. Theorem. — Every nowhere-vanishing holomorphic function f on a simply connected region Ω has a holomorphic logarithm g , so that $f = e^g$. The function g may be denoted $\log f$ and determines a branch.

Source slide 241.

9.2.6. Theorem (Equivalent global Cauchy properties). — For a region Ω , the following are equivalent:

- (A) Every closed path γ has $\text{Ind}_\gamma(z) = 0$ for $z \notin \Omega$.
- (B) $\int_\gamma f(w) dw = 0$ for every closed path and every $f \in H(\Omega)$.
- (C) Every $f \in H(\Omega)$ has a primitive $F \in H(\Omega)$.
- (D) Every $f \in H(\Omega)$ with $1/f \in H(\Omega)$ can be written $f = e^g$ for some $g \in H(\Omega)$.
- (E) Every such f can be written $f = g^2$ for some $g \in H(\Omega)$.

CHAPTER 10

Meromorphic functions and residues

MATH 503, Fall 2025. October 9, 2025. Merged PDF pages 249–275.

10.1 Meromorphic functions

Source slide 250.

10.1.1. Definition. A function f is meromorphic on an open set Ω if there is a set $A \subset \Omega$ with no limit point in Ω , such that $f \in H(\Omega \setminus A)$ and f has a pole at each point of A .

10.1.2. Remark. $A = \emptyset$ is allowed, so every holomorphic function is meromorphic. No compact subset contains infinitely many points of A ; hence A is at most countable, since open subsets of $\mathbb{C}$ are σ -compact.

Source slide 251.

10.1.3. Definition. If a is a pole and $Q(z) = \sum_{k=1}^m c_k(z-a)^{-k}$ is the principal part (so $f - Q$ is removable at a), then $\text{Res}(f; a) = c_1$. For a cycle avoiding a ,

$$\frac{1}{2\pi i} \int_{\Gamma} Q(z) dz = c_1 \text{Ind}_{\Gamma}(a).$$

Source slide 252.

10.1.4. Theorem (Residue theorem). — If f is meromorphic on Ω , A is its pole set, and Γ is a cycle in $\Omega \setminus A$ with $\text{Ind}_{\Gamma}(\alpha) = 0$ for $\alpha \notin \Omega$, then

$$\frac{1}{2\pi i} \int_{\Gamma} f(z) dz = \sum_{a \in A} \text{Res}(f; a) \text{Ind}_{\Gamma}(a).$$

Only finitely many terms are nonzero.

10.2 Argument principle and Rouché

Source slide 254–257.

10.2.1. Theorem (Argument principle and Rouché's theorem). — Let γ be a closed path in a region Ω , with zero index outside Ω , and with index 0 or 1 on $\Omega \setminus \gamma^*$. Let $\Omega_1 = \{\alpha : \text{Ind}_\gamma(\alpha) = 1\}$, and let N_f count zeros in Ω_1 with multiplicity. If $f \in H(\Omega)$ has no boundary zero, then

$$N_f = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz = \text{Ind}_{f \circ \gamma}(0).$$

If $g \in H(\Omega)$ and $|f - g| < |f|$ on γ^* , then $N_g = N_f$.

10.2.2. Remark. Rouché's theorem preserves the number of zeros when two holomorphic functions are sufficiently close on the boundary.

10.2.3. Example. For $g(z) = z^{87} + 36z^{57} + 71z^4 + z^3 - z + 1$, take $f(z) = 71z^4$. On $|z| = 1$,

$$|f - g| \leq 1 + 36 + 1 + 1 + 1 < 71 = |f|.$$

Thus g has four zeros in $|z| < 1$, counted with multiplicity.

10.3 Contour examples

Source slide 259–262.

10.3.1. Example.

$$\lim_{A \rightarrow \infty} \int_{-A}^A \frac{\sin x}{x} e^{itx} dx = \begin{cases} \pi, & |t| < 1, \\ 0, & |t| > 1, \end{cases}$$

and the value is $\pi/2$ when $t = \pm 1$. Using $2i \sin z = e^{iz} - e^{-iz}$ on the indented contour Γ_A gives

$$\int_{\Gamma_A} \frac{\sin z}{z} e^{itz} dz = \phi_A(t+1) - \phi_A(t-1), \quad \phi_A(s) = \frac{1}{2\pi i} \int_{\Gamma_A} \frac{e^{isz}}{z} dz.$$

Closing in the lower and upper half-planes gives

$$\phi_A(s) = \frac{1}{2\pi} \int_{-\pi}^0 e^{isAe^{i\theta}} d\theta, \quad \phi_A(s) = 1 - \frac{1}{2\pi} \int_0^\pi e^{isAe^{i\theta}} d\theta.$$

Since $e^{isAe^{i\theta}} = e^{-As \sin \theta}$, dominated convergence yields $\phi_A(s) \rightarrow 0$ for $s < 0$ and $\rightarrow \pi$ for $s > 0$; also $\phi_A(0) = \pi/2$.

Source slide 263–265.

10.3.2. Example.

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi.$$

For $f(z) = (1+z^2)^{-1}$, the upper semicircle encloses i and

$$f(z) = \frac{1}{2i(z-i)} - \frac{1}{2i(z+i)}, \quad \text{Res}(f; i) = \frac{1}{2i}.$$

The contour integral is π , while the arc is bounded by $M/R \rightarrow 0$.

Source slide 266–269.

10.3.3. Example. For $0 < a < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1 + e^x} dx = \frac{\pi}{\sin \pi a}.$$

Use $f(z) = e^{az}/(1 + e^z)$ on a rectangle of height 2π . The sole pole is πi , with residue $-e^{a\pi i}$. If I_R is the bottom integral and the vertical sides vanish, the top is $-e^{2\pi i a} I_R$, so

$$(1 - e^{2\pi i a})I = -2\pi i e^{a\pi i} \implies I = \frac{\pi}{\sin \pi a}.$$

Source slide 270–275.

10.3.4. Example.

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh \pi x} dx = \frac{1}{\cosh \pi \xi}.$$

Here $\cosh z = (e^z + e^{-z})/2$. Use $f(z) = e^{-2\pi i z \xi}/\cosh \pi z$ on a rectangle of height $2i$. The poles are $\alpha = i/2$ and $\beta = 3i/2$, with residues $e^{\pi \xi}/(\pi i)$ and $-e^{3\pi \xi}/(\pi i)$. The vertical sides vanish, the top is $-e^{4\pi \xi} I$, and the residue theorem gives $I = 1/\cosh(\pi \xi)$.

10.3.5. Remark. The same method gives

$$\int_{\mathbb{R}} \frac{e^{-2\pi i x \xi}}{\cosh \pi x + \cos \pi a} dx = \frac{\sin \pi a}{\sinh(2\pi \xi)}$$

in the source's stated parameter range, with $\sinh z = (e^z - e^{-z})/2$.

CHAPTER 11

Phragmen–Lindelof and harmonic analysis

MATH 503, Fall 2025. October 13, 2025. Merged PDF pages 276–303.

11.1 Phragmen–Lindelof method

Source slide 277–280.

11.1.1. Remark. The maximum modulus principle fails on unbounded sets. On $\Omega = \{x + iy : -\pi/2 < y < \pi/2\}$, $f(z) = \exp(\exp z)$ has boundary modulus 1 but $f(x) = e^{e^x} \rightarrow \infty$. Growth control replaces boundedness in Phragmen–Lindelof arguments. For example, an entire function satisfying $|f(z)| \leq 1 + |z|^{1/2}$ is constant.

11.1.2. Theorem (Strip Phragmen–Lindelof). — Let $\Omega = \{x + iy : a < x < b\}$, with f continuous and holomorphic and $|f| < B$. Set $M(x) = \sup_y |f(x + iy)|$. Then

$$M(x)^{b-a} \leq M(a)^{b-x} M(b)^{x-a}, \quad a \leq x \leq b.$$

Source slide 287.

11.1.3. Corollary. — If the preceding assumptions hold and f is nonconstant, then $|f(z)| < \max(M(a), M(b))$ in Ω .

Source slide 288.

11.1.4. Theorem (Hadamard three-circle theorem). — For $0 < R_1 < R_2$, let g be holomorphic on the annulus and $B(R) = \max_{|z|=R} |g(z)|$. Then

$$B(R)^{\log R_2 - \log R_1} \leq B(R_1)^{\log R_2 - \log R} B(R_2)^{\log R - \log R_1}.$$

Source slide 291.

11.1.5. Theorem (Strip application). — If $|\Im z| < \pi/2$, f is holomorphic and continuous there, and $|f(x + iy)| < \exp(Ae^{\alpha|x|})$ for $\alpha < 1$, while $|f(x \pm \pi i/2)| \leq 1$, then $|f(z)| \leq 1$ throughout the strip.

11.1.6. Remark. The conclusion fails at the critical growth $\alpha = 1$, as $\exp(\exp z)$ demonstrates.

11.2 Interpolation and convolution

Source slide 294–301.

11.2.1. Theorem (Riesz interpolation). — For σ -finite measure spaces $(X, \mu), (Y, \nu)$, let T be linear on finitely simple functions and suppose

$$\|Tf\|_{q_j} \leq M_j \|f\|_{p_j}, \quad j = 0, 1.$$

For $0 < \theta < 1$, define

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then $\|Tf\|_q \leq M_0^{1-\theta} M_1^\theta \|f\|_p$.

11.2.2. Remark. Finitely simple functions are dense in L^p for $p < \infty$ on σ -finite spaces, yielding a unique bounded extension.

11.2.3. Lemma (Three-lines lemma). — If F is analytic in $0 < \Re z < 1$, bounded and continuous on the closure, with $|F| \leq B_0$ on $\Re z = 0$ and $|F| \leq B_1$ on $\Re z = 1$, then $|F(z)| \leq B_0^{1-\theta} B_1^\theta$ on $\Re z = \theta$.

11.2.A. EXERCISE. Prove the three-lines lemma.

Source slide 302–303.

11.2.4. Theorem (Young's convolution inequality). — If $1 \leq p, q, r \leq \infty$ and $1 + 1/q = 1/p + 1/r$, then

$$\|f * g\|_{L^q(\mathbb{R})} \leq \|f\|_{L^p(\mathbb{R})} \|g\|_{L^r(\mathbb{R})}, \quad (f * g)(x) = \int_{\mathbb{R}} f(x-y)g(y) \, dy.$$

11.2.B. EXERCISE. For fixed $g \in L^r$, verify directly that convolution maps L^1 to L^r and $L^{r'}$ to L^∞ with the stated norms, then derive Young's inequality by interpolation.

CHAPTER 12

The Fourier transform

MATH 503, Fall 2025. October 16, 2025. Merged PDF pages 304–327.

Source slide 305–307.

12.0.1. Definition. For $a > 0$, let $\mathcal{F}_a$ be the class of functions holomorphic on $S_a = \{z : |\Im z| < a\}$ and satisfying a uniform exponential bound of the form $|f(x + iy)| \leq Ce^{-b|x|}$ on every narrower strip $|\Im z| \leq b < a$. Such bounds imply the corresponding bounds for all derivatives by Cauchy's formula.

Source slide 308–311.

12.0.2. Theorem. — If $f \in \mathcal{F}_a$, then for each $0 < b < a$ there is B such that

$$|\widehat{f}(\xi)| \leq Be^{-2\pi b|\xi|}, \quad \widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i x \xi} dx.$$

The proof shifts the real contour to $\Im z = -b$ for $\xi > 0$ and to $\Im z = b$ for $\xi < 0$; the vertical sides vanish by the exponential estimates.

Source slide 312–318.

12.0.3. Remark. The transform has rapid decay. The Fourier inversion formula is

$$f(x) = \int_{\mathbb{R}} \widehat{f}(\xi)e^{2\pi i x \xi} d\xi$$

under the source's decay hypotheses. The auxiliary identity used is

$$\int_{\mathbb{R}} \frac{\sin(A\xi)}{\xi} d\xi = \pi \operatorname{sgn}(A).$$

12.0.4. Theorem (Poisson summation). — For f in the source class,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \widehat{f}(n).$$

Source slide 323–327.

12.0.5. Remark. The Gaussian identity gives the theta transformation. For $t > 0$,

$$\vartheta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi t n^2}, \quad \vartheta(t) = t^{-1/2} \vartheta(1/t).$$

12.0.6. Theorem (Paley–Wiener type statement). — If $|\widehat{f}(\xi)| \leq Ae^{-2\pi\alpha|\xi|}$, then

$$f(z) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi z} d\xi$$

defines a holomorphic function in the strip $|\Im z| < \alpha$.

CHAPTER 13

Conformal mappings

MATH 503, Fall 2025. October 20, 2025. Merged PDF pages 328–359.

Source slide 329–331.

13.0.1. Definition. A conformal map $\Omega_1 \rightarrow \Omega_2$ is a one-to-one holomorphic map onto Ω_2 with nonzero derivative. It is an analytic homeomorphism, preserving oriented angles and infinitesimal shapes.

Source slide 332–339.

13.0.2. Theorem. — *The Cayley map*

$$F(z) = \frac{i - z}{i + z}$$

conformally maps the upper half-plane $\mathbb{H}$ onto the unit disc $\mathbb{D}$.

13.0.3. Theorem. — *For $n \geq 1$, the map $z \mapsto z^n$ conformally maps $\{0 < \arg z < \pi/n\}$ onto $\mathbb{H}$.*

13.0.4. Theorem. — *The map $z \mapsto z^2$ conformally maps the upper half-disc onto the first quadrant. The exponential maps $\mathbb{H}$ conformally onto the horizontal strip $\{0 < \Im z < \pi\}$.*

Source slide 340–350.

13.0.5. Theorem (Schwarz lemma). — *If $f \in H(\mathbb{D})$, $|f| \leq 1$, and $f(0) = 0$, then $|f(z)| \leq |z|$. If equality holds at a nonzero point, $f(z) = \lambda z$, $|\lambda| = 1$.*

13.0.6. Definition.

$$\varphi_a(z) = \frac{a - z}{1 - \bar{a}z}, \quad a \in \mathbb{D}.$$

13.0.7. Remark. The maps φ_a are disc automorphisms, satisfy $\varphi_a^{-1} = \varphi_a$, and preserve the disc. Schwarz's lemma applied after these automorphisms gives

$$|f'(w)| \leq \frac{1 - |f(w)|^2}{1 - |w|^2}.$$

13.0.8. Theorem. — *If f is an automorphism of $\mathbb{D}$ and $f(w) = 0$, then $f = \lambda \varphi_w$ for some $|\lambda| = 1$.*

Source slide 351–359.

13.0.9. Definition.

$$\mathrm{SL}_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc = 1 \right\}, \quad f_M(z) = \frac{az + b}{cz + d}.$$

13.0.10. Theorem. — $\mathrm{Aut}(\mathbb{H}) = \{f_M : M \in \mathrm{SL}_2(\mathbb{R})\}$. The Cayley map conjugates these transformations to the automorphisms of $\mathbb{D}$.

CHAPTER 14

The Riemann mapping theorem

MATH 503, Fall 2025. October 23, 2025. Merged PDF pages 360–386.

14.0.1. Theorem (Riemann mapping). — *Every simply connected region $\Omega \neq \mathbb{C}$ is conformally equivalent to $\mathbb{D}$.*

Source slide 362–375.

14.0.2. Lemma. — *Every open set $\Omega \subset \mathbb{C}$ admits compact sets $K_n \subset \Omega$ with $K_n \subset \text{int } K_{n+1}$ and $\bigcup_n K_n = \Omega$.*

14.0.3. Lemma (Hurwitz). — *If $f_n \in H(\Omega)$ converge uniformly on compact subsets to f , and each f_n is zero-free, then f is zero-free or identically zero.*

14.0.4. Definition. A family $\mathcal{F} \subset H(\Omega)$ is normal if every sequence in it has a subsequence converging uniformly on compact subsets to a holomorphic function or to infinity.

14.0.5. Theorem (Montel). — *Every locally uniformly bounded family of holomorphic functions is normal.*

Source slide 376–386.

The proof of the mapping theorem uses the family

$$\Sigma = \{\psi \in H(\Omega) : \psi \text{ one-to-one and } \psi(\Omega) \subset \mathbb{D}\},$$

normality, maximization of $|\psi'(z_0)|$, disc automorphisms, and Schwarz's lemma. The extremal map is onto $\mathbb{D}$.

CHAPTER 15

Runge's theorem and approximation

MATH 503, Fall 2025. October 27, 2025. Merged PDF pages 387–416.

Source slide 389–397.

15.0.1. Definition. A rational function is P/Q for polynomials P, Q , $Q \neq 0$. Its poles are the zeros of Q after cancellation.

15.0.2. Theorem (Cauchy theorem for compact sets). — For a compact K and an open neighborhood Ω , a finite balanced collection of oriented grid intervals gives a cycle whose index is one on K and zero outside Ω .

15.0.3. Theorem (Runge). — Let $K \subset \mathbb{C}$ be compact and choose one point α_j in each component of $\widehat{\mathbb{C}} \setminus K$. If $\Omega \supset K$ is open and $f \in H(\Omega)$, then f is uniformly approximated on K by rational functions with poles among the α_j .

Source slide 403–408.

15.0.4. Corollary (Polynomial approximation). — If $\widehat{\mathbb{C}} \setminus K$ is connected, every function holomorphic near K is uniformly approximated on K by polynomials.

15.0.5. Remark. Connectedness of the complement is necessary: holes require poles in the complementary components.

Source slide 409–411.

15.0.6. Theorem (Mittag-Leffler). — Given a discrete set of points and prescribed principal parts there, there exists a meromorphic function with exactly those principal parts.

Source slide 412–416.

15.0.7. Theorem (Characterizations of simple connectivity). — For a region Ω , the following conditions are equivalent:

- (a) Ω is conformally equivalent to $\mathbb{D}$.
- (b) Ω is simply connected.
- (c) Every holomorphic function on Ω has a primitive.
- (d) Every nonvanishing holomorphic function has a logarithm.
- (e) Every nonvanishing holomorphic function has a holomorphic square root.
- (f) Every holomorphic function near compact subsets is approximable by polynomials.

CHAPTER 16

Harmonic functions

MATH 503, Fall 2025. October 30, 2025. Merged PDF pages 417–456.

16.0.1. Theorem (Cauchy–Riemann in Wirtinger form). —

$$\partial_x f = \frac{1}{i} \partial_y f.$$

16.0.2. Definition. A real function u is harmonic when $u_{xx} + u_{yy} = 0$.

16.0.3. Theorem. — *The real and imaginary parts of a holomorphic function are harmonic. A harmonic function vanishing on a nonempty open subset of a region vanishes throughout that region.*

16.0.4. Definition. If u is harmonic, v is a harmonic conjugate when $u + iv$ is holomorphic; equivalently $u_x = v_y$ and $u_y = -v_x$.

16.0.5. Theorem. — *Every harmonic function on a disc has a harmonic conjugate. A region is simply connected exactly when every harmonic function on it has one.*

16.0.6. Definition. Continuous u has the mean-value property if

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta$$

whenever the closed disc is contained in the region.

16.0.7. Theorem. — *An MVP function satisfies the maximum principle; a nonconstant one cannot attain an interior maximum or minimum. Conversely, a continuous MVP function is harmonic.*

16.0.8. Definition.

$$P_r(\theta) = \Re \frac{1 + re^{i\theta}}{1 - re^{i\theta}} = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}.$$

16.0.9. Theorem (Dirichlet problem for a disc). — *For continuous real boundary data f on $C(a, \rho)$ there is a unique continuous harmonic u on the closed disc with $u = f$ on its boundary, given by the Poisson integral*

$$u(a + re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_{r/\rho}(\theta - \varphi) f(a + \rho e^{i\varphi}) d\varphi.$$

CHAPTER 17

Infinite products

MATH 503, Fall 2025. November 3, 2025. Merged PDF pages 457–494.

17.0.1. Theorem (Jensen). — If f is holomorphic near $\overline{D(0, R)}$, $f(0) \neq 0$, and $a_1, \dots, a_N$ are its zeros in the disc, counted with multiplicity,

$$\log |f(0)| + \sum_{j=1}^N \log \frac{R}{|a_j|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta.$$

17.0.2. Definition. An entire function has order at most ρ if $|f(z)| \leq Ae^{B|z|^\rho}$ for suitable A, B .

17.0.3. Theorem. — An entire function of order at most ρ has at most Cr^ρ zeros in $|z| \leq r$ for large r .

17.0.4. Definition. An infinite product $\prod(1 + a_n)$ converges if its partial products converge to a nonzero limit. If $\sum |a_n| < \infty$, it does.

17.0.5. Lemma. — If $|z| < 1/2$, then $|\log(1 + z)| \leq 2|z|$.

17.0.6. Theorem (Weierstrass product). — For every sequence $|a_n| \rightarrow \infty$, an entire function exists whose zeros are exactly the prescribed a_n , with prescribed multiplicities.

17.0.7. Definition.

$$E_0(z) = 1 - z, \quad E_k(z) = (1 - z) \exp \left(z + \frac{z^2}{2} + \cdots + \frac{z^k}{k} \right).$$

17.0.8. Theorem (Hadamard factorization). — An entire function of finite order factors as a polynomial times canonical factors $E_k(z/a_n)$ and an exponential factor of bounded degree.

CHAPTER 18

Gamma function and summation

MATH 503, Fall 2025. November 10, 2025. Merged PDF pages 495–549.

18.0.1. Theorem. —

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

18.0.2. Definition.

$$\Gamma(z) = \frac{e^{-\gamma z}}{z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right)^{-1} e^{z/n}.$$

18.0.3. Theorem (Euler’s functional equation). —

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(n+1) = n!.$$

18.0.4. Theorem (Reflection and duplication). —

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}, \quad \Gamma(2z)\Gamma(1/2) = 2^{2z-1}\Gamma(z)\Gamma(z+1/2).$$

18.0.5. Theorem (Integral representation). — For $\Re z > 0$, $\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt$.

18.0.6. Definition.

$$\frac{ze^{xz}}{e^z - 1} = \sum_{k=0}^{\infty} B_k(x) \frac{z^k}{k!}.$$

18.0.7. Theorem (Euler–Maclaurin). —

$$\sum_{n=a}^b f(n) = \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \sum_{r=2}^q \frac{B_r}{r!} (f^{(r-1)}(b) - f^{(r-1)}(a)) + R_q.$$

18.0.8. Theorem (Stirling). —

$$\Gamma(z) = \sqrt{2\pi} z^{z-1/2} e^{-z} (1 + O(1/z))$$

in sectors $|\arg z| \leq \pi - \delta$.

CHAPTER 19

Dirichlet series

MATH 503, Fall 2025. November 13, 2025. Merged PDF pages 550–594.

19.0.1. Definition.

$$D(s) = \sum_{n=1}^{\infty} a_n n^{-s}, \quad s = \sigma + it.$$

19.0.2. Theorem. — *There are abscissae σ_a and σ_c of absolute convergence and convergence. In every strict half-plane $\sigma > \sigma_c$, D is holomorphic and may be differentiated term by term.*

19.0.3. Lemma. — *The coefficients of a Dirichlet series are uniquely determined by the function in a half-plane of convergence.*

19.0.4. Definition. A function f is multiplicative when $f(mn) = f(m)f(n)$ for co-prime m, n .

19.0.5. Theorem (Euler product). —

$$\sum_{n \geq 1} \frac{f(n)}{n^s} = \prod_p \left(\sum_{k \geq 0} \frac{f(p^k)}{p^{ks}} \right)$$

under the source's convergence hypotheses.

19.0.6. Theorem. — *Products of Dirichlet series correspond to Dirichlet convolution of their coefficients, with absolute convergence justifying rearrangement.*

CHAPTER 20

The zeta function and the prime number theorem

MATH 503, Fall 2025. November 17, 2025. Merged PDF pages 595–631.

20.0.1. Definition.

$$\Lambda(n) = \begin{cases} \log p, & n = p^k, \\ 0, & \text{otherwise,} \end{cases} \quad \psi(x) = \sum_{n \leq x} \Lambda(n), \quad \vartheta(x) = \sum_{p \leq x} \log p.$$

20.0.2. Theorem (Chebyshev bounds). —

$$\vartheta(x) \leq x \log 4, \quad \psi(x) \asymp x.$$

20.0.3. Definition.

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1} \quad (\Re s > 1).$$

20.0.4. Theorem (Analytic continuation and functional equation). —

$$\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

extends to an entire function and satisfies $\xi(s) = \xi(1-s)$.

20.0.5. Remark. The trivial zeros are $-2, -4, -6, \dots$; ζ has no zeros for $\Re s > 1$, and the nontrivial zeros lie in $0 \leq \Re s \leq 1$.

20.0.6. Theorem (Explicit formula). —

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}) \text{the source's limiting correction.}$$

20.0.7. Theorem (Prime number theorem). —

$$\psi(x) = x + O\left(xe^{-c\sqrt{\log x}}\right).$$

20.0.8. Conjecture: Riemann hypothesis. Every nontrivial zero of ζ has real part $1/2$.

20.0.9. Theorem (Riemann–von Mangoldt). —

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T).$$

CHAPTER 21

Quantitative estimates for the prime number theorem

MATH 503, Fall 2025. November 20, 2025. Merged PDF pages 632–693.

21.0.1. Lemma. — *The logarithmic derivative ζ'/ζ admits a renormalized partial-fraction expansion in terms of the zeros, with absolute convergence in bounded vertical strips away from zeros and gamma poles.*

21.0.2. Theorem (Zero-free region). — *There is $C > 0$ such that $\zeta(s)$ has no zero*

$$\rho = \beta + i\gamma, \quad \beta \geq 1 - \frac{C}{\log(|\gamma| + 3)}.$$

21.0.3. Lemma (Perron kernel). —

$$\frac{1}{2\pi i} \int_{\kappa - iT}^{\kappa + iT} \frac{x^s}{s} ds$$

is 1 for $x > 1$, 0 for $0 < x < 1$, and has the half-value at $x = 1$, up to an error controlled by $x^\kappa/(1 + T|\log x|)$.

21.0.4. Theorem (Truncated Perron formula). — *For a Dirichlet series, the finite-height Perron integral recovers the summatory function with the explicit truncation error displayed in the source lecture.*

21.0.5. Theorem (Landau explicit formula). —

$$\psi(x) = x - \sum_{|\gamma| \leq T} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}) + \text{truncation error}.$$

21.0.6. Theorem (Korobov–Vinogradov). — *There is $C > 0$ such that no zero lies in*

$$\sigma \geq 1 - \frac{C}{(\log(|t| + 3))^{2/3} (\log \log(|t| + 3))^{1/3}}.$$

21.0.7. Remark. The source obtains its best stated PNT error term by inserting this zero-free region into Perron’s formula.